Journal of Archaeological Method and Theory (2026) 33:33
https://doi.org/10.1007/s10816-026-09768-2

RESEARCH

Small Drills, Big Questions—Investigating Late Neolithic Dickenbännli Drills Through Experimentation, Technology, Traceology and Residue Analysis

Hannah Huber[1,2] · Claus-Joachim Kind[4] · Patrick Schmidt[1,3] · Yvonne Tafelmaier[4] · Flavia Venditti[1]

Received: 6 November 2025 / Accepted: 1 February 2026
© The Author(s) 2026

Abstract

The manufacture of limestone beads using so-called Dickenbännli drills has been proposed to be part of a large-scale societal development: specialization of labor and the resulting emergence of social stratification. These small lithic drills are abundant at many Middle and Late Neolithic sites in Southern Germany and Northern Switzerland, frequently co-occurring with limestone beads. However, the use of the drills for bead manufacture is mostly assumed rather than demonstrated. Furthermore, there is little experimental, technological and traceological evidence to support the hypothesis of a specialized production, except for the site of Hornstaad-Hörnle, where the argument is based on spatial distribution and manufacturing skill. The Late Neolithic assemblage of Kohlhau-Abri, which includes both drills and disc bead fragments, serves as a case study to reconstruct the manufacturing process of drills and limestone beads and evaluate their interconnection, ultimately examining the technical skills involved. This was achieved through explorative and controlled experiments, coupled with traceological and residue analysis of both drills and beads. Our use wear results show that the drills were used, likely to work limestone. Raman spectroscopy of a white residue found at the tip of one of the drills indicates calcite, the primary component of limestone. Neither the experimental nor the archaeological evidence suggests a necessity for specialists in bead manufacturing. Our study highlights how the application of multiple analytical approaches can provide us with complementary lines of evidence, and how such a holistic approach enables us to tackle not only small-scale but also large-scale questions.

Keywords Kohlhau-Abri · Use wear analysis · Chaîne opératoire · Specialization · Young Neolithic

Extended author information available on the last page of the article

Published online: 25 February 2026

Introduction

Specialization of labor has long been linked to the emergence of social stratification within a society (Marx, 1867; Rousseau, 1754). The earliest instances of this economic process are often sought during the Neolithic (*e.g.* Baysal, 2013; Kerner, 2010; Thomas, 2009; Wright, 2014). In Southern Germany and Northern Switzerland, Early Neolithic Linear Banded Pottery settlement systems (LBK) are traditionally viewed as subsistence economies with minimal specialized labor and an egalitarian social structure (Krauß *et al.*, 2020; Lüning, 1988). In contrast, the manufacture of Early Bronze Age metal objects supposedly exhibit a higher degree of specialization than previously recognized crafts, and bronze items are often regarded as markers of prestige in a vertically stratified society (Altorfer *et al.*, 2016; Hansen, 2018). The narrative of specialized metalworking leading to more "complex" or stratified societies has long accompanied discussions of early metals (*e.g.* Childe, 1950; recently Hansen, 2018; Klimscha, 2020; Renfrew, 1978).

For the Neolithic in Southern Germany and Northern Switzerland, multiple activities and crafts have been proposed and discussed in the context of specialization processes that may have contributed to stratification processes. Chert mining, which increases in scale in the Middle Neolithic, has been proposed to be the work of specialists (Engelhardt & Binsteiner, 1988; Willms, 1982), although it has been more recently argued that specialization is not required (Roth, 2008). During the Late Neolithic, there are many proposed candidates for specialization: the production of ground and knapped axes in ateliers (Ewald & Sedelmaier, 1994; Speck, 1988), procurance of yew wood (Favre & Jacomet, 1998), as well as the production of wooden vessels (Müller-Beck, 1991) and arrow points (Hoffstadt, 2005; Lüning & Zürn, 1977). Furthermore, the production of limestone beads and the so-called Dickenbännli drills have been suggested to be the work of specialists (Heumüller, 2009; Hoffstadt, 2005).

These beads, both tubular and disc-shaped, as well as the small lithic drills assumed to have been used in their production, are abundant at various Middle and Late Neolithic sites in the region. Comprehensive reviews of these drills and limestone beads (as well as other ornaments) in a broad spatiotemporal context are available by Hoffstadt (2005) and Heumüller (2009) respectively. Drills and tubular beads co-occur e.g. at Hornstaad-Hörnle IA with nearly 19,000 drills and 3800 beads (Heumüller, 2009; Hoffstadt, 2005), as well as at other Lake Constance sites, such as Markelfingen-Kleine Espen, Wahlwies-Betten and Bodman-Löchle (Schlichtherle, 1990). At Offwiesen (Dieckmann *et al.*, 1998) and Galeriehöhle II and III (Nadler, 1985), the drills are associated with disc-beads.

While there is no shortage of either material in the record, the drills and their use are not well understood. Analysis of the lithic materials is often limited to typological assignments, despite the category Dickenbännli drill not reflecting a type in the traditional sense. D'Aujourd'hui (1976) classified multiple types within the category and considered them reflective of particular functions and working of different materials, and describes Dickenbännli drills as small,

irregularly shaped objects with a steeply retouched drill tip and a more or less pronounced base. Hoffstadt (2012) argues that these types are not clearly separable from each other and adds more elongated, needle-like drills to the subgroupings of the category Dickenbännli drills. Detailed metric data is published for one assemblage with such drills (Kind, 1989). For Hornstaad-Hörnle, the lithic materials (Hoffstadt, 2005), ornaments, and their production waste (Heumüller, 2009) have been presented, and a use wear study on the Hornstaad-Hörnle drills was conducted by A. Pawlik (Pawlik, 2000), the results of which are published only in summary. Schlichtherle (1990) linked drills and beads through comparing drill tip dimensions and perforation diameters, but as of yet, no study has combined the analysis of drills and beads from a traceological and residue perspective.

Furthermore, the manufacturing process of drills, as well as beads, is not well understood from an experimental perspective. At Hornstaad-Hörnle, materials reflective of the entire drill and bead *chaîne opératoire* have been found. The spatial distribution of these finds being limited to particular buildings and specific areas of the settlement is the main argument for specialization. Seeberger's (1992) perforation experiments, which feature a fixed bow drill construction that allows for 2200 rotations per minute, produce an immense output of beads, leading Hoffstadt (2005) to suggest surplus production at Hornstaad-Hörnle—which would be consistent with an economic structure that could contribute to the emergence of hierarchies. Heumüller (2009) criticizes the assumptions made in the experiments and disagrees with the interpretation of surplus production. She does, however, support the argument for specialized production of both drills and beads, and suggests that specific fine motor skills as well as a certain level of experience are required to drill limestone, necessitating some degree of specialization (Heumüller, 2009: 116).

There is a clear need for more data on multiple fronts, especially given the scope of the economic processes to which the craft of drill- and beadmaking has been tied. The aim of this study was twofold. On the one hand, we evaluated the skills involved in drill- and beadmaking through experiments, and in doing so, tested the hypothesis that particular fine motor drilling skills necessitated specialization of this craft. On the other hand, the Late Neolithic assemblage of Kohlhau-Abri serves as a case study. By analyzing drills and beads found at the site, we aimed to connect the drills and beads through use wear and residue evidence to improve our understanding of their manufacturing process. For this, we conducted experiments to examine the formation of traces in beads and drills, and produced reference pieces for analysis of the archaeological artefacts. We applied both a low- and high-power approach to the archaeological sample, as well as Raman spectroscopy to characterize residues. Technological analysis of the archaeological materials was conducted concurrently with the analyses presented herein, enabling insights from the technological analysis of the archaeological drills to be incorporated into the experimental program. In turn, the results of this work informed technological interpretations, enabling a more holistic understanding of the materials.

Archaeological Setting and Assemblage

Kohlhau-Abri is a rockshelter in the Swabian Jura, located near Bissingen ob Lontal in the periphery of the Lone valley. It was discovered and excavated by the State Office for Cultural Heritage Baden-Württemberg between 2015 and 2018. A preliminary report presenting the site and its single finds has been published (Kind & Beutelspacher, 2020), and detailed analyses of the various materials are underway. A total of 19 m^2 was excavated, with the vast majority of finds stemming from a core area of 10 m^2, which was excavated to a depth of 3 m. All finds recovered during excavation were plotted in three dimensions, and all removed sediments were waterscreened.

Seven archaeological horizons (AH) were defined in total *i.e.* continuous horizons of find scatters in the excavated area. AH 3 is the second most numerous in finds, followed by the underlying AH 4, which is assigned to the Mesolithic. AH 3 is well separated from the younger AH 2, as the two find concentrations are 15 cm apart in height, and the sediments in between contain only a few finds. The scatters of AH 4 are 5 to 10 cm lower than those of AH 3, and in some areas, it was not possible to assign finds to one or the other horizon. Finds from AH 3 include lithic artefacts, faunal materials, as well as ceramic sherds, some of which can be assigned to the Late Neolithic. The three limestone bead fragments from AH3 were found during waterscreening. An agglomeration of charcoal, burnt clay spots and burned archaeological materials has been interpreted as a hearth feature. ^{14}C-dating was done at the Curt-Engelhorn-Zentrum für Archäometrie in Mannheim. A bovid pelvis fragment assigned to AH 3 was dated to 3972–3825 calBC, and a wild boar metapodium yielded a date of 4588–4506 calBC. The younger date corresponds to Late and the older date to Middle Neolithic. The lithic materials were considered Late Neolithic, due to the presence of small lithic drills classified as Dickenbännli drills (Kind & Beutelspacher, 2020). The faunal materials mostly consist of juvenile wild pig. They are highly fragmented and display root etching. Possible cutmarks have been identified. The results of the analyses of both macro- and microfauna of the site are forthcoming.

Analysis of the lithic assemblage of AH 3 was conducted at the same time as this study. The technological analysis of the drills and the assemblage will be the subject of a separate paper. The results indicate a palimpsestic assemblage characterized by import and export, with little to no on-site blank production. Multiple instances of recycling could be recorded, likely also making use of materials left on the surface in previous occupations. The drills were likely produced on-site as indicated by semi-finished drills in the assemblage as well as artefacts recycled into drills. A total of 90 drills and drill fragments were identified in the assemblage, all made from Jurassic chert (Examples: Fig. 1). The drills share few technological similarities. Several morphologies appeared multiple times, such as drills with a centered tip vs. a left-leaning tip, but the respective artefacts lack technological or metric regularities that could justify a techno-typological distinction. Drill production is better characterized as expedient rather than curated, using any blank, flake or blade, whether complete or fragmented, cortical or not.

Fig. 1 Examples of drills from the Kohlhau assemblage, in which the active parts of the drill tips and the passive, prehensile area of the drills are marked

Many overall morphological differences were determined by this unstandardized blank selection, as they translate to different cross sections, side profiles, in the overall tool as well as the drill's tip. Length of the drills (measured along the main axis following the drill tip) ranged from 8.5 to 29.9 mm, with a median of 14.7 mm. Drill tip dimensions, as measured 1 mm away from the very end of the tip, appeared more standardized, with a mean width and thickness of 1.8 mm.

The Archaeological Drill Sample

All 90 drills identified in the AH 3 assemblage were screened for residues and presorted based on preservation. Fragmented drills, those with extensive metal scratches from waterscreening, and those with thermal alterations, such as potlid fractures obscuring traces, were excluded as candidates for use-wear analysis. Of the 90 drills, sixteen were "well-preserved" according to these criteria, and fourteen exhibited tiny, whitish, amorphous accumulations at their tips. Of the 16 well-preserved drills, those without residues were selected for use-wear analysis ($N = 11$), while the 5 remaining drills with white residues were not microscopically analyzed to avoid losing information and to enable future residue analysis. Nonetheless, one drill with the largest amount of residue was analyzed using Raman spectroscopy but was not assessed for use-wear. All archaeological materials were exclusively handled while wearing powder-free nitrile gloves.

For residue screening, an Olympus SZX7 stereomicroscope with magnification ranging from 8 × to 56 × and equipped with a LED ring light source was used. Drills with potential residues were separated.

Raman spectroscopy was chosen to identify the chemical composition of potential residues, as it is a non-destructive, no-contact, and time-efficient method of analysis that allows for the detection of mineral as well as organic compounds. Raman spectra of both the residue as well as the chert constituting the drill were obtained, using a Renishaw InVia Reflex Raman Microscope, which is a dispersive Raman spectroscope with a green exciting laser (532 nm), and a 20 × magnification objective. The spectral range was 200–1200 cm^{-1}. For each sample, 5 repetitions were recorded, for a total acquisition time of 7–15 min. The software Spectragryph was used to process

the Raman spectra. The spectra were then compared to references of different materials available in the literature (Bouchard & Smith, 2003; Dufresne *et al.*, 2018). Reference spectra for calcite and quartz were obtained from the RRUFF database, IDs R050048 and R110104 (Lafuente *et al.*, 2015).

The drills selected for use wear analysis were rinsed under tap water and gently cleaned with a neutral detergent (Derquim®). They were then placed in a neutral detergent solution and put in the ultrasonic bath for 5 min. After rinsing, the drills were again put in the ultrasonic bath for 5 min in distilled water. Any remaining grease on edges and ridges was removed as needed, using cotton swabs dipped in ethanol.

The focus of the traceological analysis was on the active area identified in the explorative experimentation. However, the passive area was also examined to assess the overall preservation of the artefact's edges, ridges, and surface, and affirm any observed traces at the active area as use-related when they are absent from the rest of the artefact.

The 11 selected drills were described and photographed using both a low-power (< 56 × magnification) and a high-power microscope (up to 500 × magnification) (see Keeley, 1980; Odell & Odell-Vereecken, 1980). The drills were examined using the Olympus SZX7 stereomicroscope for description of macro traces. Photographs at this scale were taken using a Hirox-2500 digital microscope. For high-power-analysis, an Olympus BX53M widefield microscope with vertical incident light was used for both description and documentation. Terminology and description of the macro- and micro-traces followed Claud *et al*. (2019) and included distribution, morphology, orientation, initiation and termination of removals, and rounding. For polish, distribution, contours, micro-topography and density were noted (see Claud *et al.*, 2019; Fischer *et al.*, 1984; Ho Ho Committee, 1979; Plisson, 1985). Rounding was classified into light, medium and pronounced rounding. Light rounding refers to rounding that is barely recognizable macroscopically. Rounding was considered medium when it was clearly visible under low power, and obscured and changed features such as edges, removals and ridges. Pronounced rounding was such that an original feature had been completely smoothed over. Left edge, right edge, distal end and the dorsal ridges were documented separately. The edges of the drills are referred to in dorsal view *i.e*. "left dorsal" (LD) and "left ventral" (LV) refer to the same edge.

The three archaeological limestone bead fragments were examined for technological traces using the Olympus SZX7. Photographs were taken with the Hirox-2500.

The use wear analysis and residue screening and documentation were performed at the Material Culture Laboratory (hereafter MCL) at the University of Tübingen. The Raman spectroscopy was performed at the Competence Center for Archaeometry Baden-Württemberg at the University of Tübingen.

The Experimental Program

The experiments were performed at the MCL and conducted in two rounds. Explorative or first-level (Marreiros *et al.*, 2020) experiments were conducted to assess the minimal degree of skill that is needed to use drills for perforating limestone and thereby test the hypothesis of specialization being required. Additionally, these

experiments served to gain general insights into variables affecting use and wear formation, which influenced the design of a second round of experiments. The second round of experiments was more controlled, to document the development of use wear in the drills and produce experimental references for the analysis of the archaeological materials. Additionally, to evaluate bead output per drill based on previous experiments, notes on efficiency were recorded in this round. The first author conducted the experiments. She had no previous experience with grinding stone, and her use of lithic drills was limited to perforating animal teeth.

The materials used in the experiments all stemmed from southwestern Germany. Local sandstones available at the MCL were used for grinding. The limestone was collected directly at the site of Kohlhau-Abri in the form of small, slab-like fragments of weathered limestone that require little working to fashion bead preforms. Limestone is abundant on the forest floor of the surrounding area, as the rock shelter itself consists of limestone. The Jurassic cherts from the Swabian Jura which were used to make the experimental drills were provided by R. Walter, who also produced the drills by knapping. The blanks from which the drills were made were preselected based on their thickness matching the dimensions of the archaeological specimen.

Both handheld and hafted drilling were performed in the experiments to ascertain whether the different modes produce qualitatively different traces in drills or beads (see Gurova et, &al., 2014), and to assess the minimum degree of skills needed to perforate limestone. While it is often assumed that drills in the Neolithic were used hafted, in pump or bow drills, and previous works only included hafted drilling (Seeberger, 1992; Semenov, 1964), there is no direct evidence for this in the German Neolithic to the knowledge of the authors, either through finds of hafts at the pile dwellings around Lake Constance, or through hafting traces on drills. A simple clamping haft was chosen as a hafting mode, since no potential adhesive residues were observed in the Kohlhau-Abri assemblage, and previous investigations of Dickenbännli drills did not find adhesive residues either, despite excellent preservation at Hornstaad-Hörnle (Pawlik, 2000).

The basic workflow of the bead production process for both experiments was based on finds at comparative sites with semi-finished beads and beadmaking waste products. At Gaimersheim, finds of limestone raw material, preforms and waste products suggest the following *chaîne opératoire:* disc beads were made from secondary deposits of limestone, in this case from river gravels. They appear to have been shattered into debris, which was ground into slabs with roughly parallel surfaces, and then worked into disc preforms with facets on the outer surface. The preforms were perforated biconically, and a smooth, rounded shape was achieved through a final step of grinding (Weinig, 1987). The finds at Hornstaad-Hörnle suggest a similar workflow for tubular limestone beads, with mostly local glacial limestone gravels having been used as a raw material (Heumüller, 2009).

For the explorative experiments, the experimental drills were made to resemble various drill morphologies and dimensions represented in the assemblage to evaluate whether these differences influence handling *e.g.* in the case of the drill tip being centered along the main axis of the tool versus being off-center. Different motions were tested, as well as the addition of water as a lubricant in both grinding and

drilling. Notes on required skills, handling and ease of use were taken during these experiments.

In the second, controlled round, seven drills with centered tips and similar dimensions were used. Four of these drills were used for set times and documented in between, at 3 min, 10 min and 30 min, to track the development of traces with prolonged use. One of these drills was hafted. The remaining three sample drills were used until a decrease in perforation efficiency was observed, then resharpened, and used for another 30 min to enable comparison with the other four drills. Also here, one of the drills was hafted (see Table 1). Back-and-forth strokes (back-and-forth = 1 stroke, see Tringham *et al.* (1974)) were counted for the first minute of use in three drills to assess the average amount of strokes per minute. The duration needed to perforate the bead preforms was also documented. Grinding was performed with adding water as a lubricant, and drilling was performed dry.

The experimental reference drills (ExDs) from the second round were described and photographed in the MCL using the same workflow as described above for the archaeological sample. First, they were described and documented in their unused state. Comparing the pictures of the unused and the different used states of the ExD's allowed for tracking changes in the active area. Following use and before documentation, the ExDs were cleaned and from this point onward only handled while wearing powder-free nitrile gloves. The drills were first rinsed under tap water, then placed in a solution with 3% acetic acid for 10 min, and rinsed again. Next, neutral soap was used to clean the drills. After another water rinse, the drills were left in a 3% natron solution for 10 min, rinsed, cleaned with neutral soap and rinsed again. Finally, they were put in distilled water in an ultrasonic bath for 5 min. Cotton swabs dipped in ethanol were used as needed to remove leftover grease on edges and dorsal ridges.

Bead blanks and beads from both rounds of experimentation were described and recorded at different stages of production, using the Olympus SZX7 and the Hirox-2500 microscopes. Unperforated bead blanks, perforated blanks, as well as finished limestone beads were examined for technological traces resulting from the performed actions.

Results

The Experimental Program

In the explorative experiments, 8 limestone beads were produced in 5 h. The involved activities and motions did not appear to go beyond standard grinding and drilling. Grinding the already slab-like limestone into bead preforms was possible in both dry and wet conditions. However, it was much easier to grind using water, as the water and the removed material mixed together and likely facilitated abrasion. In drilling, however, this was not as convenient because it made it hard to see the perforation depth.

Making round preforms from ground-down slabs of ca. 2.5 mm thickness was in part possible by breaking the slab into little pieces by hand, with minimal grinding

Table 1 Overview of use steps in the second round of experiments

Drill	Mode of drilling	Resharpening	Use steps
ExD3	Handheld	Yes	Until efficiency has notably decreased, resharpening, then another 30 min of use
ExD4	Handheld	Yes	Until efficiency has notably decreased, resharpening, then another 30 min of use
ExD7	Handheld	Yes	Until efficiency has notably decreased, resharpening, 30 min of use
ExD5	Handheld	-	3–10–30 min use
ExD8	Handheld	-	3–10–30 min use
ExD2	Hafted	-	3–10–30 min use
ExD6	Hafted	Yes	Until efficiency has notably decreased, resharpening, then another 30 min of use

of the edges being required. In preforms from 2.5 to 2.8 mm thickness, the slabs could be broken by hand using an edge (of a different rock) as support, and a little more grinding was needed to produce roughly circular preforms compared to thinner preforms. Drilling the limestone bead preforms was easy, whether done by hand or with a hafted drill. Different overall dimensions of the drills did not have an observable effect on handling. Relatively little pressure was required to hold them in place, either between the index finger and a resting thumb or by the clamping haft. The different drill morphologies did not affect handling either. The clamping haft held them secure in all cases. In handheld use, it was easy to use the drills when the drill tip and thumb were aligned along the rotational axis of the wrist. Used as such, the different drill shapes, sizes, and the drill tip morphologies did not affect handling. Other alignments of the drill tip and the wrist were attempted but proved too difficult and uncomfortable to be likely preferred over the more intuitive option. It was not necessary to fix the bead preforms to perforate them. In handheld drilling, it was possible to hold them in the hand not used for drilling, while in hafted use, placing them on a piece of leather provided sufficient static friction for the bead not to move. Finishing up the edges of the bead preforms turned out to be the most difficult part of the production process. As the beads became smaller, it was more difficult to hold them and grind the beads' edges.

Duration of the individual steps in the manufacturing process is difficult to assess, particularly for the various grinding steps, as there are multiple uncontrolled variables *e.g.* dimensions and porosity of the raw material, applied force, *etc*. To provide a rough estimate: grinding one of the collected pieces down to a slab that can be broken into beads required about an hour of work. The number of preforms made from one ground slab varied, depending on the size of the slab. Full, biconical perforation of a preform took 23 min on average, although the recorded time varied greatly. The longest perforation took 65 min, the shortest only 7 min. The shortest perforation time was achieved in hafted use, although an 8 min and a 10 min perforation were achieved in handheld use as well.

In the more controlled second round of experiments, use of the drills produced observable traces in all cases, in short (3 min) as well as long (30 min) use times. Macroscopic traces (removals and rounding) were present in all used drill stages. Regarding traces at the micro-level, polish could be recorded in eight stages of use. No striations were identified. Drilling speed during the experiments was, on average, 59 strokes per minute. Rounding was the most frequent trace. Localized at the distal tip end, rounding was observed to different degrees in all use cases (Table 2, Table 3). Edge rounding and rounding of the dorsal ridge were observed in all cases except for the 3-min use stage of ExD8 (see Table 2, Table 3, Fig. 2). Unfortunately, edge rounding prevented the identification of removal initiations, and often obscured the shape, termination and distribution of removals. The degree of rounding is generally increased with use time, and it increased faster in the hafted drill (see Fig. 3).

Edge damage was also frequent; use produced dorsal or ventral edge removals in all drills except for ExD5's 30 min time step (Table 2), where traces seemed to have stabilized (Fig. 4). All edge removals were classified as very small (< 0.5 mm). Save for a few unclear cases at the distal end, removals were perpendicular to the edge. Removals at the edges and dorsal ridges were most commonly

Table 2 Damage to the ExDs edges in the active area. *LD/RD/DD*, left/right/distal dorsal; *LV/RV/DV*, left/right/distal ventral. Hafted drills are underlined. 1 st use refers to the flexible use times of the drills that were used until a subjective decrease in efficiency was noted

	Irregular	Lunate	Semi-circular	NA	Tri-angu- lar	Qua-drang-ular	Edge rounding
ExD2, 3 min	LD	—	LD	—	—	—	Light
ExD5, 3 min	LD, LV, RV, DV	RV	RD, LV, DD	—	—	—	Light
ExD8, 3 min	LD, LV	LV	—	—	—	—	—
ExD2, 10 min	LD, RD, LV, RV, DD, DV	—	—	—	—	—	Light
ExD5, 10 min	DD	—	RV	LD, RD	DD	—	Light
ExD8, 10 min	RD	—	LV	LD	—	—	Medium
ExD3, 1 st use (33 min)	LD, RD, LV, RV, DD	RV	RD	—	—	—	Medium
ExD4, 1 st use (66 min)	RD, LV	—	RD, LV, DV	LD, DD, DV	—	—	Pronounced
ExD6, 1 st use (29 min)	RD, LV, DV	—	RV	LD	—	LV	Pronounced
ExD7, 1 st use (53 min)	RD, RV	LV, RV	—	LD, DV	—	LV	Pronounced
ExD2, 30 min	LV, DD, DV	—	—	LD, RD	—	—	Medium
ExD3, 30 min	RD	—	—		—	—	Medium
ExD4, 30 min	RV	—	DV, RV, LV	DV	—	—	Light
ExD5, 30 min (no new damages)	—	—	—	—	—	—	Medium
ExD6, 30 min	LD, RD, RV	LV			—	—	Light
ExD7, 30 min	LD, RD			DD, DV	—	—	Light
ExD8, 30 min	LV, DV	RV	RV	LD, RD	—	—	Medium

Table 3 Presence/absence of selected traces in the ExDs. Hafted drills are underlined. 1 st use refers to the flexible use times of the drills that were used until a subjective decrease in efficiency was noted

	Ridge damage	Ridge rounding	Ventral distal damage (beveled tip)	Distal damage (crushing)	Distal rounding
ExD2, 3 min	Yes	Medium	—	—	Medium
ExD5, 3 min	—	Light	—	—	Medium
ExD8, 3 min	Yes	—	—	—	Medium
ExD2, 10 min	—	Pronounced	Yes	—	Pronounced
ExD5, 10 min	Yes	Light	Yes	Yes	Medium
ExD8, 10 min	Yes	Medium	—	Yes	Medium
ExD3, 1 st use (33 min)	Yes	Medium	Yes	—	Pronounced
ExD4, 1 st use (66 min)	Yes	Pronounced	—	Yes	Pronounced
ExD6, 1 st use (29 min)	Yes	Pronounced	—	—	Pronounced
ExD7, 1 st use (53 min)	Yes	Pronounced	Yes	—	Pronounced
ExD2, 30 min	—	Medium	—	—	Medium
ExD3, 30 min	Yes	Medium	—	—	Pronounced
ExD4, 30 min	Yes	Pronounced	—	—	Pronounced
ExD5, 30 min	—	Medium	—	—	Pronounced
ExD6, 30 min	—	Light	Yes	—	Pronounced
ExD7, 30 min	Yes	Light	—	—	Medium
ExD8, 30 min	Yes	Medium	—	Yes	Medium

irregular in shape with step terminations. While irregular in overall shape, removals were generally wider than long. Edge removals were observed dorsally as well as ventrally. Lunates and semi-circular removals, typically with feathered terminations, were also common. They were usually located on the ventral side at the edges, and in the case of lunates, exclusively so. A special case of ventral distal damage was defined. In some cases, a removal parallel to the drill's main axis removed the ventral part of the tip, creating a new "beveled" tip.

Where statements on distribution could be made, a discontinuous distribution of isolated removals along the edges was the most common mode, closely followed by discontinuous superimposed removals and continuous superimposed removals. Aligned removals were comparatively uncommon.

Polish (see Table 4) was recorded in nine use cases and appeared uniform in the used drills. It was recorded ventrally, at the left and right edges of the active area as well as the distal tip end, and in three out of the eight occurrences, on dorsal ridges. The polish consisted of marginal, tiny spots or thin lines of polish directly on the edge or ridge. Contours were very sharp, and the polish appeared very smooth and flat (see Fig. 5). It was possible to observe the development of

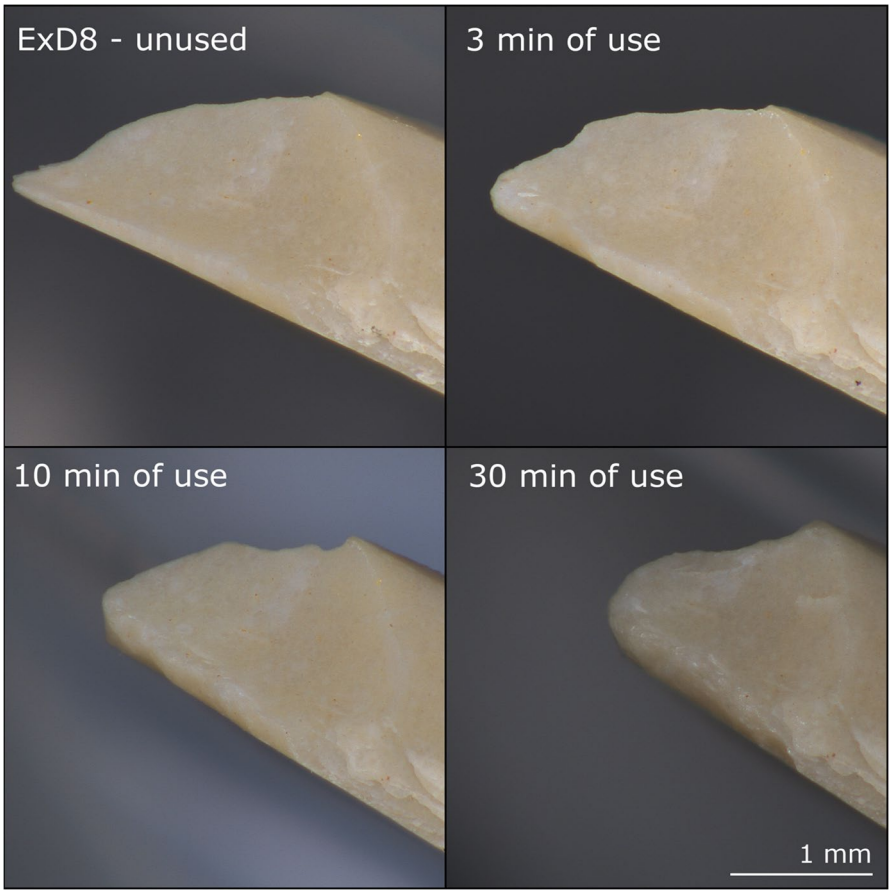

Fig. 2 Development of traces in ExD8, left lateral view of the tip. Pictures shown were taken on the same tool and at the same spot after successive periods of use. The increase in distal rounding and ridge removals is especially apparent

polish in one case; in ExD8, the polish identified at 10 min of use could still be located after 30 min of use. The polish line extended along the right ventral edge of the drill, but not onto the ventral surface. In all other cases, the area in which polish had formed had been removed within the next use of the tool by edge removals, rounding or resharpening.

Polish was also recorded on two drills in an unused state, more specifically, a resharpened state. The drills were documented before as well as immediately after resharpening. An antler tool was used to resharpen the drills through pressure retouch. This resulted in a polish, which, similar to the polish resulting from use, was located at the ventral edge, although it extended further onto the surface and it was slightly more diffuse and rougher in texture than the polish observed in the used drills (Fig. 5).

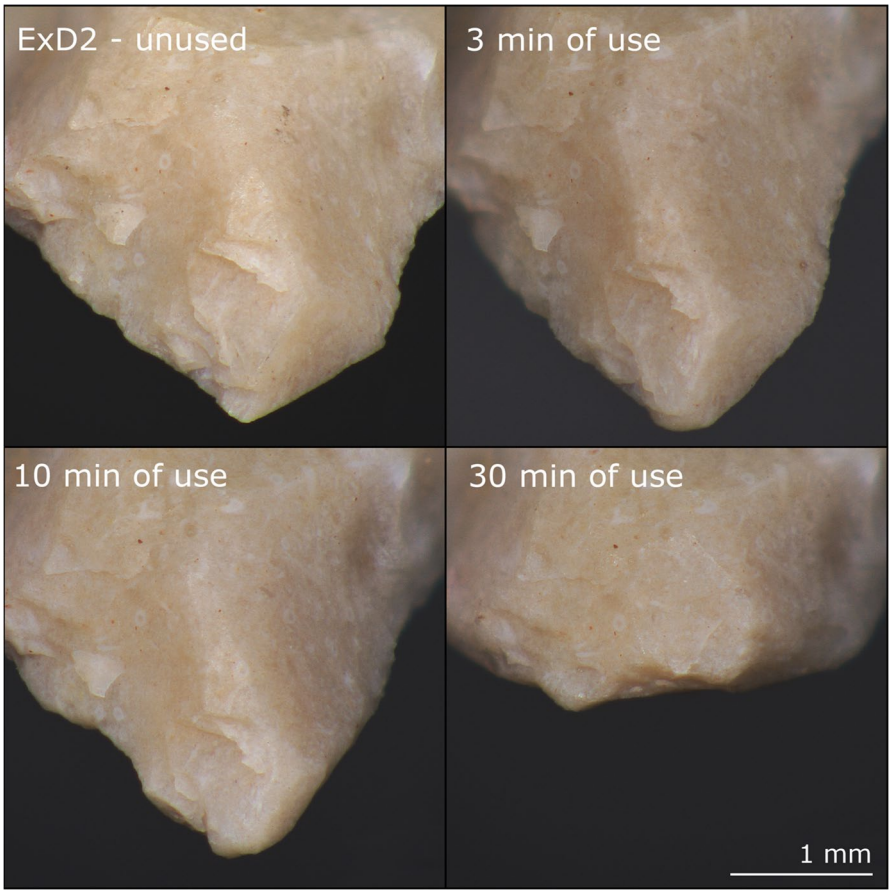

Fig. 3 Development of traces in ExD2 at the dorsal tip after successive periods of use, especially distal and ridge rounding. Pictures shown were taken on the same tool and at the same spot

Before resharpening, ExD3 had the earliest and most noticeable change in efficiency. The second perforation of the first bead took longer to initiate and complete. For ExD6 and ExD7, a second bead was perforated in its entirety. A steady but slower decrease in efficiency was noted. ExD4 was very efficient initially, and even more so on the second perforation of the first bead. While its efficiency was reduced in the second bead, again, efficiency was better for the second perforation. Thus, a third bead was initiated. However, after 7 more minutes of drilling, virtually no perforation had been achieved, and the drill was then also resharpened. Figure 6 shows the change in time requirements for 1 mm perforation depth between the first and the second perforation of a bead, calculated based on perforation depth and measured time.

The identification of production traces in the experimental bead blanks, semi-finished beads, as well as finished beads, was possible in all cases. All preforms

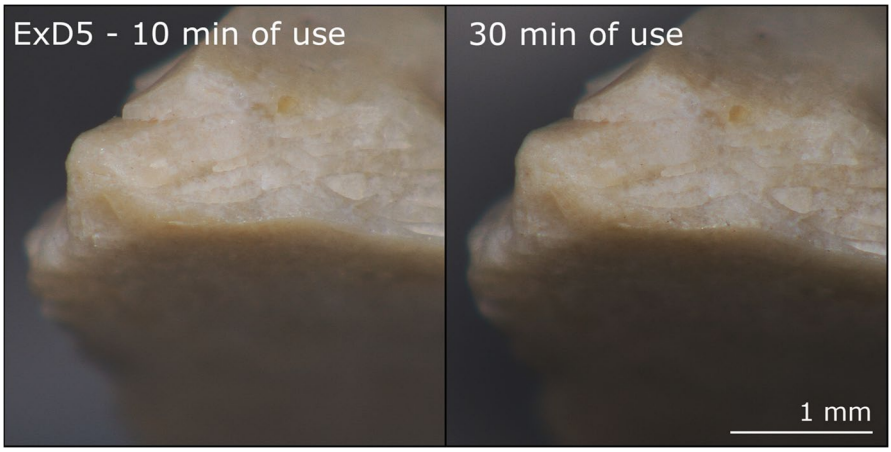

Fig. 4 Stabilized traces in ExD5. The pictures were taken on the same tool at the same spot, with 20 min of use in between, producing little to no observable changes in use traces

Table 4 Location of observed polish in the ExDs. Hafted drills are underlined. 1 st use refers to the flexible use times of the drills that were used until a subjective decrease in efficiency was noted

	Polish on ventral tip	Polish on ventral edge	Polish on ridge
ExD2, 3 min	Yes	—	
ExD5, 3 min	—	—	Yes
ExD8, 3 min	—	—	—
ExD2, 10 min	Yes	—	—
ExD5, 10 min	—	—	—
ExD8, 10 min	Yes	Yes, right edge	—
ExD3, 1 st use (33 min)	Yes	—	—
ExD4, 1 st use (66 min)	Yes	—	Yes
ExD6, 1 st use (29 min)	—	Yes, left edge	Yes
ExD7, 1 st use (53 min)	Yes	Yes, sleft edge	—
ExD2, 30 min	—	—	—
ExD3, 30 min	—	—	—
ExD4, 30 min	—	—	—
ExD5, 30 min	—	—	—
ExD6, 30 min	—	—	—
ExD7, 30 min	—	—	—
ExD8, 30 min	Yes	Yes, right edge	—

and perforated beads displayed striations on the curved outer surfaces, the flat surface and inside of the perforation. Striations on the surfaces were of similar depth, roughly parallel and easy to recognize. In the concentric drilling striations, depth varied slightly between different areas of the perforation. Within one perforation

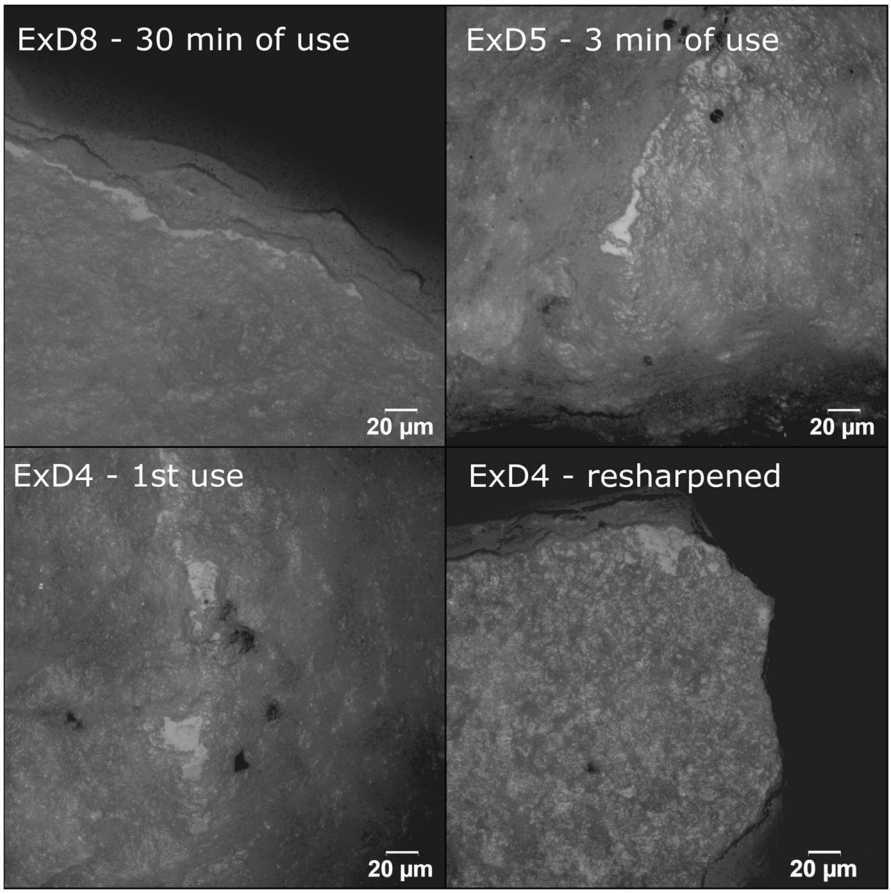

Fig. 5 Limestone polish in different locations—on ExD8, at the ventral edge, ExD5 on the central dorsal ridge of the tip, and on ExD4 at the ventral edge of the tip. Bottom right shows the polish from resharpening with an antler tool

cone, striations were parallel to each other. As for the regularity of the perforations, there was no observable difference between beads which had been perforated using handheld or hafted drills. While the most irregular perforation shapes were made with handheld drills, some very circular and regular perforations were achieved with a handheld drill as well.

The Archaeological Sample

Cleaning of the 11 drills on which use wear analysis was performed removed most sediments. Some yellow fine-grained material remained stuck in crevices, mostly of the passive area, but not to an extent where it would hinder microscopic observation. The overall preservation of surfaces and edges was good. The surfaces displayed no patination, no irregular striations and many edges of the passive area of the drills

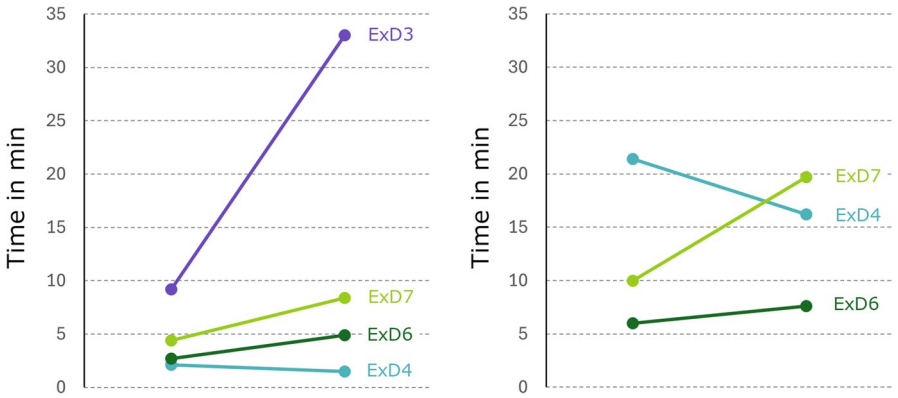

Fig. 6 Development of the time needed to perforate 1 mm of limestone on the same bead. Left: Difference between the perforation speed on the first cone and the second cone of the first bead. Right: Difference between the first and the second cone of the second bead. ExD6 was used hafted; the others were used handheld

appeared undamaged. In drill 4433_88.2, some of the ventral edge damage in the passive area appeared "fresher" *i.e.* brighter in color and less rounded, and the edges of the proximal transversal breaks on drills 4432_210.1, and 4532_80.1 appeared "fresher" as well. The majority of drills displayed thin, disorganized metal scratches in isolated spots. The metal traces of 4332_165 are wider and more extensive.

Edge damage was observed in both the active and passive areas of all analyzed drills. All dorsal removals in the active area as well as all removals in the passive area were classified as very small (<0.5 mm). For ventral removals in the active area, some reached up to 1.5 mm in size, although most were smaller than 0.5 mm as well. Where the directionality of a removal in the active area was recognizable, it was perpendicular to its respective edges. Light rounding of the edges was noted in all but one of the drills, occurring in either the passive or the active area. In six out of the 11 analyzed drills, both areas exhibited rounding (Table 5, Table 6).

Although the passive areas of the drills were not examined in detail, some wear patterns were noted to compare them to the active area. Removals in the passive area were most commonly irregular in shape, but lunate, semi-circular, and quadrangular removals occurred as well. Feathered terminations were most frequent and occurred in irregular, lunate and semi-circular removal shapes (Table 5). Where statements on distribution could be made for the passive area of the drills, removals were mostly discontinuously distributed and isolated.

In the active area, nearly all removal shapes were irregular, with lunates, semi-circular and trapezoidal removals occurring only on one edge of one drill, respectively. Regarding terminations, stepped was by far the most common, specifically in combination with irregular removal shapes. Feathered terminations occurred in individual cases, and one hinged termination of one trapezoidal removal was observed. Regarding distribution, a continuous distribution of overlapping and superimposed removals along the edges was most frequent in the active area.

Table 5 Damage to the drills' edges in the active and passive areas. *LD/RD/DD*, left/right/distal dorsal; *LV/RV/DD*, left/right/distal ventral

Drill	Area	Irregular	Lun-ate	Semi-circular	NA	Trape-zoid	Quad-rang-ular	Edge rounding
4433_88.2	Passive	X	X	—	—	—	—	Light
	Active	LD, RD, LV, DD, DV	—	—	—	RV	—	—
4332_165	Passive	X	X	—	—	—	—	Light
	Active	LD	LD	—	RD, LV, RV, DD, DV	—	—	Light to medium
4432_325	Passive	X	—	—	—	—	—	—
	Active	LD, RD, DD	—	DV	LV, RV	—	—	—
4433_223.2	Passive	X	—	—	—	—	—	Light
	Active	LD, RD, RV, DV	—	—	LV	RV	—	—
4333_101.3	Passive	X	—	—	—	—	—	Light
	Active	LV, RV, DV	—	—	LD, RD, DD	—	—	Light
4433_118.2	Passive	X	—	X	—	—	X	Light
	Active	LD, RD	—	—	DD	—	—	Light
4432_210.1	Passive	—	—	—	—	—	—	Light
	Active	LD, LV, RV	—	—	RD, DD	—	—	Light
4332_211.1	Passive	—	—	—	—	—	—	Medium
	Active	RD	—	—	DD	—	—	—
4433_223.1	Passive	X	—	—	—	—	—	Light
	Active	LD	—	—	RD, DV	—	—	Light
4433_247.7	Passive	X	X	—	—	—	—	—
	Active	LD, RD, LV, RV, DV, DD	—	—	—	—	—	Light
4532_80.1	Passive	X	—	—	—	—	—	Light
	Active	LD, RD	—	—	DD	—	—	Light

Table 6 Presence/absence of criteria of use as determined in the experimental program in the analyzed archaeological material. *LD/RD/DD*, left/right/distal dorsal; *LV/RV/DD*, left/right/distal ventral; *PA*, passive area

Drill	Ridge da-mage	Ridge rounding	Ventral distal dam-age (beveling)	Ventral edge remo-vals	Distal damage (crushing)	Distal rounding	Polish
4433_88.2	Yes	Medium	Yes	LV, RV	Possible rounded	Light	DD, DV, PA
4332_165	Yes	Medium	—	LV, RV	Yes	Medium	—
4432_325	—	Light	Yes	LV, RV	Yes	Light to Medium	—
4433_223.2	Yes	Light	Yes	RV	Yes	Light	PA
4333_101.3	Yes	Medium	—	—	Possible rounded	Light	—
4433_118.2	Yes	Medium	—	—	Possible rounded	Pronounced	PA
4432_210.1	Yes	Medium	—	LV, RV	—	Medium	PA
4332_211.1	Yes	Medium	—	—	Possible rounded	Medium	LD, RV, DD
4433_223.1	—	Light	Yes	—	Yes	Medium	PA
4433_247.7	Yes	Light	Yes	LV, RV	—	Light	—
4532_80.1	Yes	Light	—	—	Yes	Light	—

Nine out of 11 of the archaeological drills displayed both dorsal ridge damage and ridge rounding, with the remaining two displaying one or the other (Table 6, Fig. 7, Fig. 8). Beveled tips were noted for five drills, and six drills possessed ventral edge removals (Fig. 9, Fig. 10). Crushing was clearly identifiable in five cases, and in four, it appeared likely, but was obscured by distal rounding. Distal rounding occurred in all drills (Table 6).

Polish was recorded on six drills—twice in the active area, while in five drills, spots of polish were identified in the passive area, mostly on dorsal ridges and on edges (Fig. 11, Table 7). For the polish in the active area, its location originated at ventral edges, or was located on a rounded tip or a ridge. The recorded polish

Fig. 7 Rounding of the dorsal ridge and the tip on an experimental drill and archaeological example 4433_118.2

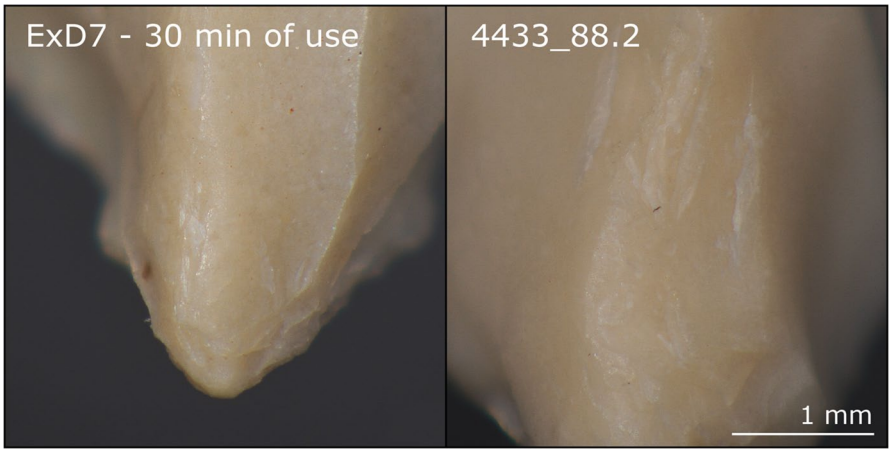

Fig. 8 Damaged dorsal ridges in experimental example ExD7 and archaeological example 4433_88.2

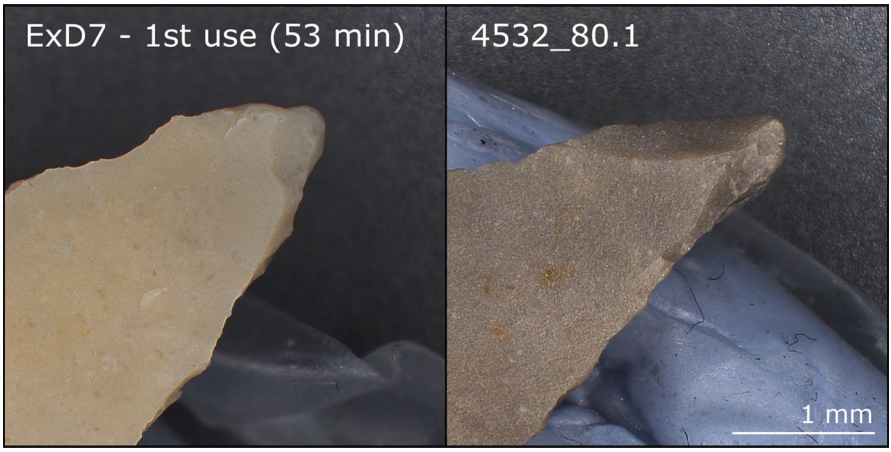

Fig. 9 Ventral removals initiating at the distal tip end ("beveled" tip), in experimental example ExD7 after 53 min of use and archaeological example 4532_80.1

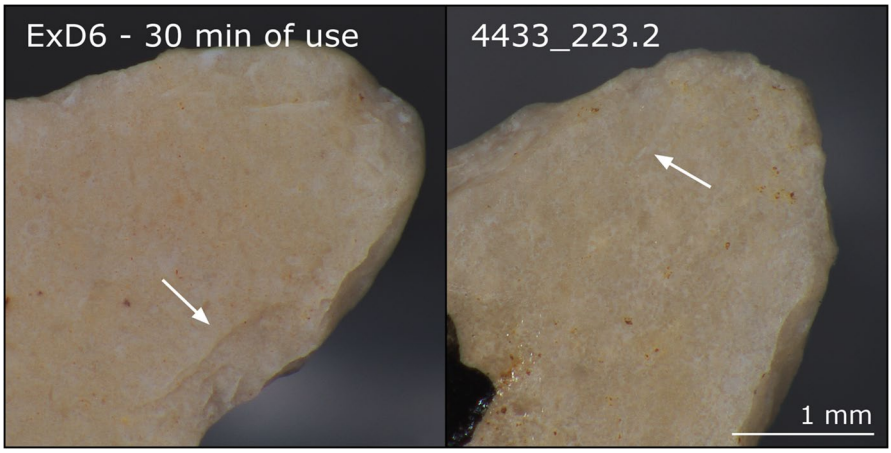

Fig. 10 Ventral edge removals on experimental example and archaeological example 4433_223.2

appeared uniform in all instances, as bright, smooth, marginal spots or lines with sharp contours and a flat topography.

The three bead fragments show clear biconical perforation, and because they are fragmented, the perforation can be seen in cross-section (Fig. 12). Traces in the form of striations were observed in two of the three bead fragments, 327.1 and 245.1 (Fig. 13). The striations on the surfaces are all of similar depth and run parallel to each other. In the perforations, concentric striations vary slightly more in depth.

In 245.1, parallel striations were present all around the curved outer bead surface. Here, as well as in 327.1, individual striations appear to "spill over" onto the flat surface in a few spots where the two surfaces meet. Concentric striations

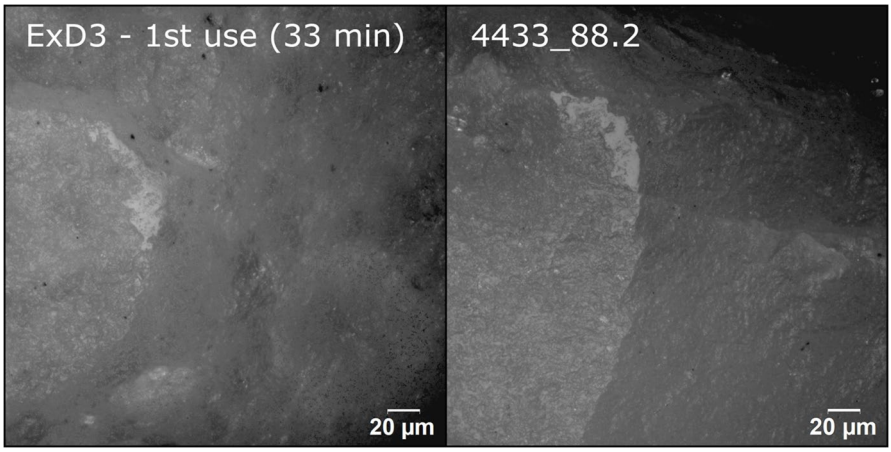

Fig. 11 Ventral polish on the tips of experimental example ExD3 and archaeological example 4433_88.2. The tip is oriented to the right

Table 7 Location of observed polish in the archaeological drills

Drill	Polish on ventral tip	Polish on ventral edge	Polish on ridge	Polish in passive area
4433_88.2	Yes	—	Yes, on rounded tip	Yes, on ventral edge
4332_165	—	—	—	—
4432_325	—	—	—	—
4433_223.2	—	—	—	Yes, on dorsal ridge
4333_101.3	—	—	—	—
4433_118.2	—	—	—	Yes, on dorsal ridge
4432_210.1	—	—	—	Yes, on dorsal ridge
4332_211.1	—	Yes, on right edge	Yes, on rounded tip and ridge	—
4433_223.1	—	—	—	Yes, on dorsal ridge
4433_247.7	—	—	—	—
4532_80.1	—	—	—	—

inside of the perforation of 245.1 could also be recorded, more densely concentrated at the outer edge of the perforation cones. No striations or other traces could be observed on the flat surface of the bead, which appeared smooth. In 327.1, striations were recorded on the curved outer surface and one of the flat surfaces. The striations are clearly visible around the curved outer surface and almost throughout the entire perforation, except for a small area where the two cones meet. On one of the flat surfaces of the bead, isolated striations that are parallel to each other are visible (Fig. 13). The remaining bead, 73.1, bore no traces and appeared smooth all over. The edges between the outer and the flat surface appear rounded as well as the edges of the perforation.

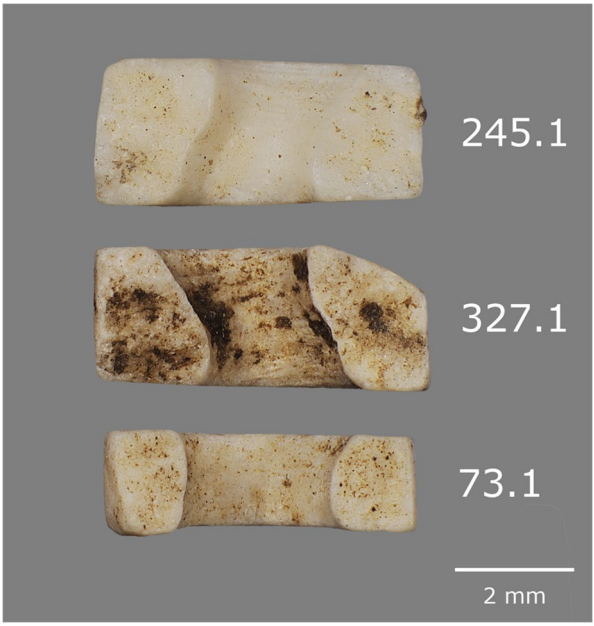

Fig. 12 Lateral view of the fragmented beads, showing biconical perforation

Regarding residues, all 90 of the archaeological specimens displayed sediment adhesions, for which it was possible to relate the material to the sediment descriptions of the different geological horizons. Most common were a very fine-grained, yellow material, a grey-brown sand, and a light beige material. Sediment was concentrated in concavities and crevices of the drill. In 14 drills, a white residue was observed (Fig. 14). This residue was only ever present in the active areas of the drill, within 1 mm of the distal tip end. It is lodged in the surface depressions and crevices of the tip. The white residue is consistent in all 14 drills and appears as a fine-grained, homogenous and opaque material.

Raman spectroscopy was applied to one drill (4632_534.1) in which the most residue was preserved (Fig. 14). The remaining 13 drills were not analyzed with Raman spectroscopy due to the respective residue locations and amounts making it unlikely to obtain reliable measurements.

In the Raman spectrum obtained from the white residue on 4632_534.1, two distinct bands are recognized. There is one strong band at 1087 cm^{-1}, and another weak band at 283 cm^{-1} (see Fig. 15). These bands are consistent with calcite (CaCO3). In the chert spectrum, one weak band was distinguishable at 466 cm^{-1}. This band is characteristic of quartz, as in chert (Schmidt *et al.*, 2012). Reference spectra of quartz (SiO2, RRUFF-ID R110104) show the strongest band at 464 cm^{-1}. The 1087 cm^{-1} and 283 cm^{-1} calcite bands were absent from the chert measurement.

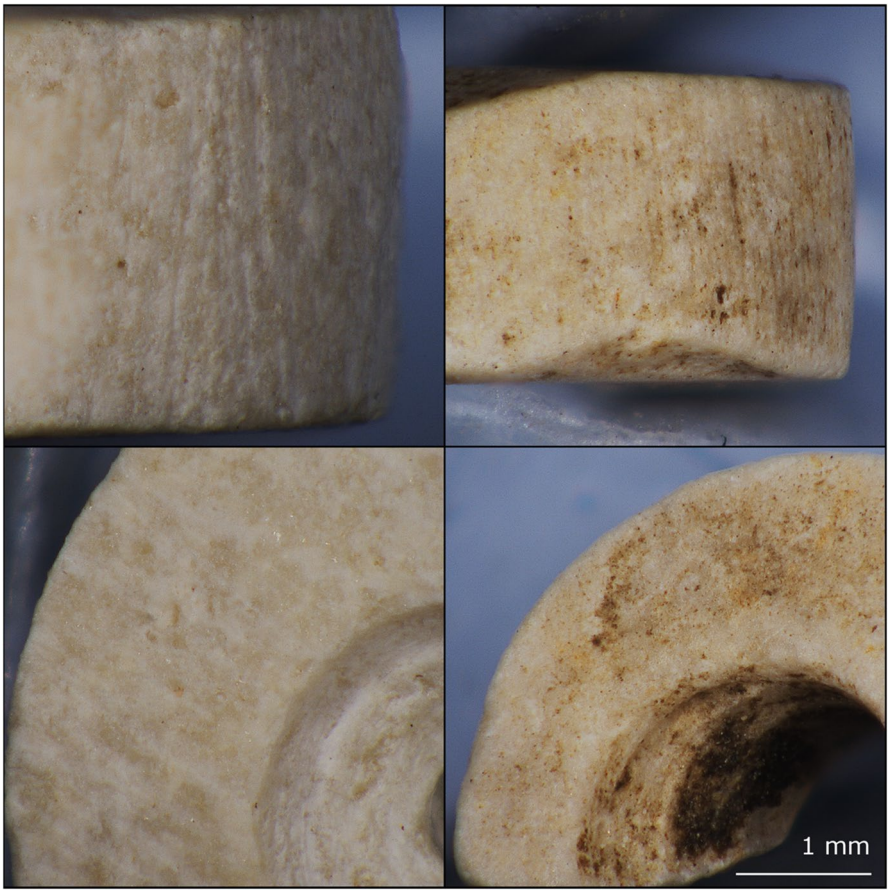

Fig. 13 Striations on an experimentally produced bead (left) and on archaeological bead 327.1 (right)

Discussion

Use of the Kohlhau Drills

The traces identified in the experimental drills align closely with those found in archaeological specimens. In the experimental material, we observed significant damage and rounding at the edges, as well as on the dorsal ridges and the distal tip, both dorsally and ventrally. These specific wear patterns were also present in the archaeological samples, suggesting a consistent use for the Kohlhau drills. The observed traces, especially rounding of the tip and the edges, have also been reported by previous experimental work on drills (*e.g.* Calley & Grace, 1988; Coşkunsu, 2009; Gurova *et al.*, 2014).

However, several of the traces documented in our experimental reference drills appear ambiguous when examined within an archaeological context. The variability

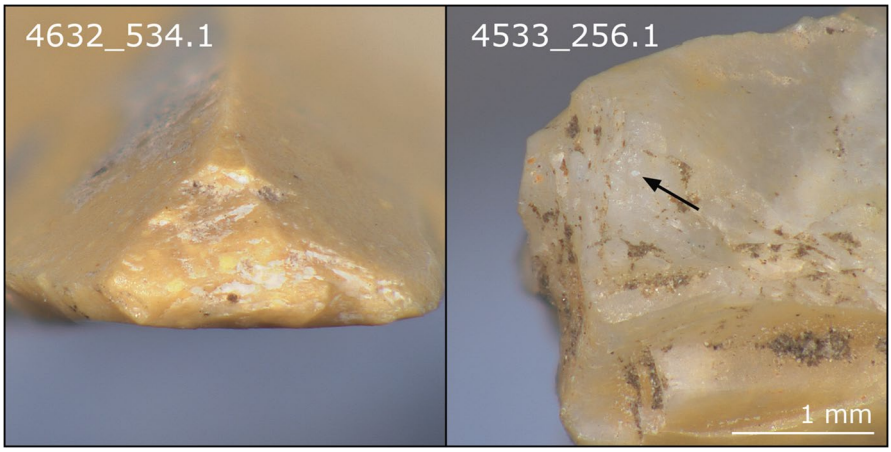

Fig. 14 Drill with the most amount of white residue, 4632_534.1, and drill 4533_256.1, in which the least amount of white residue is preserved

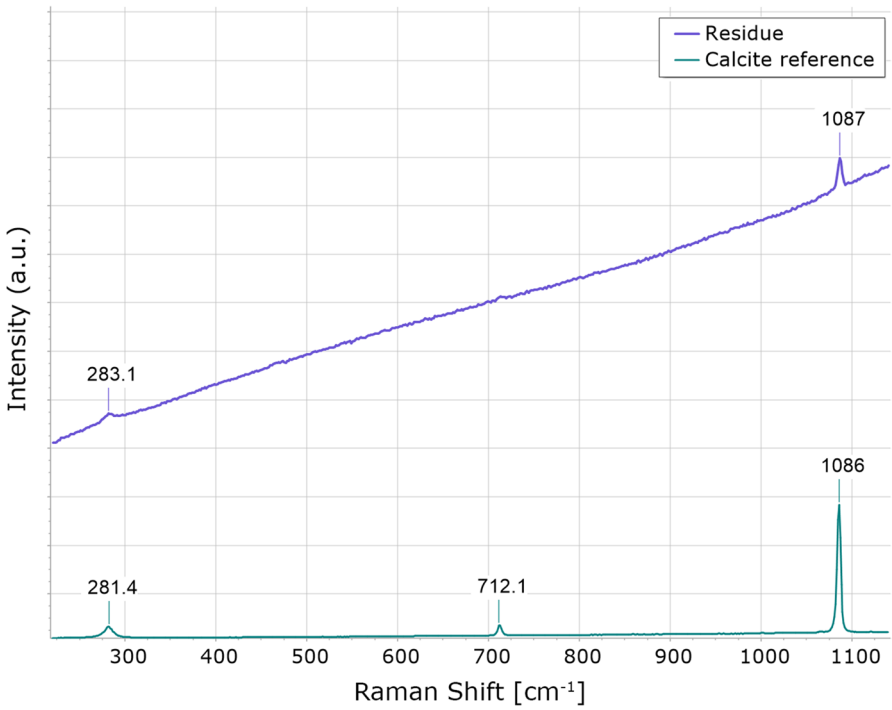

Fig. 15 Raman spectra of the white residue and a calcite reference

in preservation conditions and the lack of a pre-use comparison mean that not all usage traces may be readily identifiable as such. Dorsal edge removals were some of the earliest and most frequent use traces, but the irregularly shaped removals with

step terminations recorded after use closely resemble the retouch-related removals that are also present in the unused drills. Ventral edge removals could be ambiguous as well; some of the drills in the assemblage feature alternating retouch *i.e*. dorsal retouch on one, ventral retouch on the other edge of the drill tip, meaning it is not possible to exclude that ventral removals were the result of intentional retouch. However, distal ventral removals—"beveled" tips—initiate at the very point of the drill tip. Producing such a removal through retouch would be more difficult to achieve. It is thus more likely to be related to use.

The smooth, marginal polish that occurred in used experimental drills holds the potential to qualitatively identify use as well as determine the worked material. While action also determines the appearance and distribution of polish, a correlation between different polishes and worked material is generally accepted (Aoyama, 1995; González-Urquijo & Ibáñez-Estévez, 2003; Keeley, 1980; Yamada & Sawada, 1993). Van Gijn's (1990) experiments indicate that different materials can be distinguished in drilling. While drill polish is often present in isolated spots and appears smooth and matt regardless of worked material, different topographies were noted between wood, bone, antler, and shell. This is supported by the observations herein. While at first glance, the polish observed in the resharpened drills before use is similar to the one observed after use—marginal, smooth and bright—a difference in topography can be noted. The retouch polish, produced by an antler tool, was domed, whereas the limestone use polish was flat. This flat topography is a possible feature of a distinct limestone polish. Polish associated with cutting limestone (Van Gijn, 1990) appears flat, as well as a polish on an archaeological drill reported as a limestone polish (Coşkunsu, 2009) (although use is very intensive in this case). Use wear studies related to the engraving of limestone also report brightness and flat topography (Garate *et al.*, 2023; López-Tascón *et al.*, 2020). Thus, drawing on the available literature and the observations made herein, a marginal, bright polish in isolated spots or lines with sharp contours, and a flat topography, is considered indicative of limestone working.

As such, polish recorded in the active areas of the drills is a strong argument for use. The presence of limestone fragments in the sediment, however, is a possible complication. Artefacts could, through major movements of sediment or through excavation methods such as waterscreening, brush up against limestone. This could conceivably lead to the formation of polish, creating false positives for limestone working. Whether postdepositional movement of sediment or movement of artefact against sediment and other objects during waterscreening can cause polish to form should be investigated in future work.

The most secure use-related traces are rounding and ridge damage (see Fig. 16). Distal rounding was present in all ExD-use stages, edge and ridge rounding in all but one, and ridge damage was observed in five out of 17 use cases. There was no used drill without either ridge rounding or ridge damage. Given that preservation generally allows for the identification of traces, rounding and ridge damage are identifiable in archaeological material, and they are not produced by the manufacturing process of the tool. Even when the overall artefact may have been rounded by postdepositional processes, different degrees of rounding in certain areas can attest to use. Rounding and ridge damage can thus be considered primary criteria

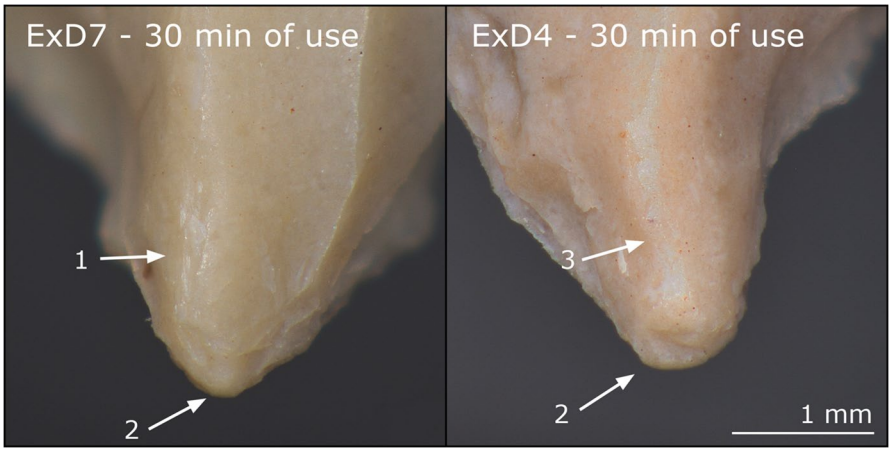

Fig. 16 Primary criteria for identifying use of a drill: bilateral removals on central dorsal ridges of the tip (1), as well as rounding of the tip (2) and the dorsal tip ridges (3)

for identifying use of an archaeological drill, supported by secondary, potentially ambiguous criteria such as polish, beveling and ventral edge removals.

Based on the applied criteria, all analyzed archaeological drills were used, exhibiting either ridge rounding or ridge damage. Distal rounding was recorded in all analyzed drills as well. In nine out of the 11 drills, the primary criteria were supported by at least one secondary criterion.

While light rounding of edges also occurred in the passive areas of most drills, which may create false positives concerning ridge rounding, the respective circumstances alleviate this potential problem. In 4432_325, no ridge damage and only light ridge rounding were recorded. However, this drill did not display rounding anywhere in its passive area. The light ridge rounding as a use-related alteration is supported by light and in parts medium rounding of the tip. In 4433_223.1, while there is light rounding of the edges in the passive area, ridge rounding is supported by medium distal rounding, and a beveled tip as a secondary criterion.

In the archaeological drills, crushing was notably frequent, as opposed to the experimental drills. The difference in frequency may have been caused by different degrees of applied force.

While edge damage does not appear particularly useful to identify use in drills due to the removals' similarities with retouch-related removals, one difference between experimental and archaeological drills in the distribution of edge removals should be noted. In the experimental material, fewer continuous superimposed removals are present, and continuous aligned removals were observed, which were not recorded for the archaeological material. This can be explained by the availability of before and after pictures for the experimental drills, which allowed for the identification of continuous aligned removals superimposed on the retouch removals. In the absence of knowledge about the unused state, these removals would likely be mistaken for the continuous superimposed removals that are more commonly recorded in the archaeological material.

As for the edge removals observed in the passive area of the archaeological drills, while removal shapes are similar in the passive and active areas of the same drill, terminations and distribution differ. In the passive area, feathered terminations are more common and steps less so, whereas the opposite is the case in the active area. Regarding distribution, the much higher frequency of discontinuous isolated removals in the passive area is striking, indicating that the edge removals of the passive area were produced by a process other than use.

As for the polish recorded in the passive area of the drills, a use-related or possibly taphonomic/modern origin is not apparent from appearance alone. The polishes recorded in the passive as well as the active areas look the same, indicating that they result from contact with the same material. Based on the polish's features, this material is likely limestone. As previously mentioned, we cannot currently exclude that limestone polishes could have also been formed by postdepositional processes or excavation techniques such as waterscreening. Since no major movement of sediments or materials is apparent at the site (see Kind & Beutelspacher, 2020), and all drills with polish in their passive areas were recovered during waterscreening, the latter would be the more probable case. The thin and disorganized metal traces observed in the archaeological drills, possibly caused by the metal sieves, demonstrate a certain amount of friction enacted on the drills during excavation and find processing. In the case of 4433_223.1, such metal traces surround the dorsal ridge polish in the passive area. A modern origin of some of the polish is also suggested by metal traces near the polish on a dorsal ridge of 4433_223.2, as well as the location of the polish on 4432_210.1 on the edge of the fresher-looking, transversal basal break.

Alternatively, these polish spots may be related to transport, prehension or hafting. The traces in the passive area were not the focus of this work. The inclusion of handheld versus hafted was done to assess skill and handling, and to determine whether these different modes produce qualitatively different use traces or technological traces in the beads, which they do not seem to do. However, it should be noted that the distribution of the passive area polish is consistent with the distribution of hafting traces on drills described by Rots (2008). Feathered removals in the passive area could indicate prehensive wear (Rots, 2004). For the site Trimmbach-Dickenbännli, hafting of these drills has been suggested, but not formally investigated (d'Aujourd'hui,). At Hornstaad-Hörnle, hafting traces were found on lithic tools in general, but not on drills (Pawlik, 2000). Further research on the prehension and hafting traces developed in the passive area of drills made of Jurassic chert could provide valuable insights into how these tools were utilized.

The more ambiguous origins of passive area polish do not, however, mean dismissing polish in the active area. The appearance and location of the active area polish observed in the archaeological drills closely match the polish in the experimental drills. Especially in the cases of 4332_211.1, it is highly unlikely that multiple spots of polish could have formed during waterscreening, but only within the very small active area, and nowhere in the much larger passive area of the drill. Additionally, unlike some of the passive area polishes listed above, none of the active area polish spots were associated with metal traces. This suggests that the active area polish results from the use of the archaeological drills on limestone.

Furthermore, the results from the residue analysis indicate that the white residue is limestone. The observed bands in the spectrum of the white residue at the tip of 4632_534.1 are indicative of calcite, the primary component of limestone. In the chert spectrum, only quartz can be identified, while the calcite bands are absent.

The calcite residue at the drill tip of 4632_534.1 is most probably caused by human activity, as the material appears to have become lodged in the surface depressions and crevices of the drill from someone working the limestone with it. It is unlikely that the residue could have been deposited through other processes, such as moving against the limestone in the sediment or having been moved against limestone during waterscreening due to the residue's location. It was only present right at the tip in 4632_534.1. In a taphonomic or excavation-related accidental deposition of the residue, some of it would be expected in the far larger passive area of the drills. However, it was only present in the area which would have come into contact with limestone during drilling, in 4632_534.1 as well as the other 13 drills in which this white material could be observed.

One trace that is commonly reported in drills in other studies but is absent in our observations should be mentioned: circular striations at the tip (Calley & Grace, 1988; Gurova *et al.*, 2014; Viola *et al.*, 2017). This is likely related to either much longer use times in these cases, a higher number of rotations due to hafted use, or even a combination of both.

Regarding the beads, the observed traces on the curved outer surface as well as the striations inside the biconical perforations in beads 327.1 and 245.1 correlate well with the traces in the experimental material, suggesting that the beads' preforms were shaped by grinding and the perforations were the result of drilling. Similar striation traces are commonly observed in stone beads from other contexts, in experimental work as well as in the archaeological materials (Bains, 2012; Gurova *et al.*, 2014; Viel, 2021; Wright *et al.*, 2008).

The near absence of striations on the archaeological beads' flat surfaces, however, does not correlate with the striations on the experimental material. Taphonomic processes being responsible for removing striations on the flat surfaces is unlikely due to the otherwise good preservation of striations on the same beads. Therefore, the flat bead surfaces cannot be assumed to have been produced using the same process as in the experiments. It is possible that an untested method, such as finishing the beads' edges while they are strung (see *e.g.* Wright *et al.*, 2008), could have generated enough friction to remove striations on the flat surfaces of the beads. Scraping could have also produced a smooth surface with few striations (Bonnardin, 2004). Alternatively, wearing the beads might have erased the traces. After all, the three beads are fragments and could have been left at the site because they broke after being worn for some time.

The lack of any striation traces in bead 73.1 is notable—here too, wear mechanisms might be responsible. Since the beads were found close to each other (neighboring quadrants), and the other two beads show well-preserved striations, taphonomy is unlikely to be the cause. If the bead was worn over a long period of time, movement on a string, against other beads, against clothing or skin could have removed all existing striations and rounded its edges. This hypothesis should be tested in future experiments on the use wear of limestone beads.

Based on the three bead fragments found at Kohlhau, it is not feasible to make more detailed statements on the manufacturing techniques employed in the manufacture of beads. Analysis of a larger assemblage of beads, at a site which includes semi-finished beads, preforms and production waste, combined with experiments tailored to the archaeological materials could shed light on specific questions regarding grinding technique in general, or the individual steps such as finishing the beads. Metric data on disc bead dimensions could provide hints regarding standardization and possibly related manufacturing techniques. Promising candidates are other proposed workshop sites, such as Galeriehöhle II and III (Nadler, 1985) or Gaimersheim (Weinig, 1987), or for a settlement context, Altdorf Hochrückacker (Piller, 2025).

The Work of Specialists?

The presence of use wear in the analyzed drills, including polish that suggests the working of limestone, and the identification of a white residue found at the tip of a drill as calcite, shows that these drills were most likely used in the production of limestone disc beads. Having established this link between the two find groups, we can address the big question: Were the people who used these drills to make beads specialists of their craft?

The hypothesis posed by Heumüller (2009)—that specific fine motor skills and previous experience are necessary for drilling and thus require some specialization—can be rejected. We found none of the skills involved in the production of limestone disc beads to be beyond what someone with little to no previous experience can learn by themselves in a short span of time. While there may be some differences in the production of the mostly tubular beads at Hornstaad-Hörnle and the disc beads at Kohlhau-Abri *e.g.* in preform making, or in fixing the bead for making the perforation using a hafted drill, use of the small drills themselves cannot be considered necessarily the task of specialists.

The low-level skills involved in beadmaking align with the results of the technological analysis of the drills and the AH 3 assemblage (manuscript in preparation), which shows drill production to be expedient as opposed to curated, with a low investment of time, effort or knowledge (see Vaquero & Romagnoli, 2018). Examining raw material procurement offers insights into effort and time investment. The Jurassic chert used for making the drills is locally available in the Swabian Jura. Additionally, no specific target blanks were employed. Instead, artefacts were created and recycled into drills using blanks with different properties. This suggests that it may not have been necessary to acquire new chert, but instead, waste material and leftovers from other reduction sequences were utilized. Consequently, the effort and time needed are relatively low, as retouching any blank can produce a drill, which does not require much time or specialized knowledge.

Some might now argue that making limestone beads could still have been the work of specialists, even though no specific expert skills were required. Indeed, at Hornstaad-Hörnle, the spatial distributions of beads, bead preforms, and drills at certain housing structures can convincingly be interpreted as the house of a

specialist craftsperson. The activities at Kohlhau-Abri and other sites believed to be disc-bead workshops (*e.g.*Nadler, 1985; Weinig, 1987) do not and cannot show such spatial patterns, as they are rockshelter and cave sites. However, in this context, it is important to consider the theoretical implications of specialized labor regarding the supposed rise of elites.

Specialized labor as it relates to social stratification is specialization in which the means of production are not collectively owned (Marx, 1867)—but specialization could, in contrast, also refer to a division of labor according to each person's abilities and needs (see Marx, 1891). It is rarely made explicit whether one considers "specialization" the former or the latter type. Discussing surplus production and commercial trade of Late Neolithic limestone beads (Hoffstadt, 2005; Maier, 1962; Seidel, 2004) indicates that the former is referred to. Thus, the question becomes whether the means of limestone bead production could have been restricted to a group of specialist craftspeople.

The physical means include raw materials for both beads and drills. At Gaimersheim and Hornstaad-Hörnle, local limestone was used (Heumüller, 2009; Weinig, 1987). At Kohlhau-Abri, there is no direct evidence for the use of local material, but it is abundant at the site and likely could not have been restricted in the region, as the Swabian Jura is a limestone plateau. Outcrops of Jurassic chert are harder to find, so this raw material might be more readily restricted. However, at Kohlhau-Abri, recycling and the probable use of waste products from other reduction sequences suggest easy access to chert objects suitable for making drills. As established, the skills needed for beadmaking—grinding, retouch, and drilling—can be learned quickly. Restricting these basic skills does not seem feasible during Neolithic daily life. Lithics remained a common everyday item, used in various activities, offering many chances to observe and learn techniques such as retouch and drilling. The same applies to grinding—even if ground stone tools were initially made in ateliers (Ewald & Sedelmaier, 1994; Speck, 1988), axes and adzes need regular resharpening during use. There could have been social restrictions on who was allowed to make beads, as well as intangible rules involving rituals tied to their symbolic meanings. However, this is difficult, if not impossible, to demonstrate archaeologically. Therefore, the simplest explanation is that it was not possible to restrict access to the materials and skills needed for limestone bead production to a small group of specialized artisans.

In consequence, the informative value of this craft regarding specialization as it relates to stratification processes is limited. Whether the beads were actually regularly produced by a specialized craftsperson is not so relevant in this context, as well as any actual employed techniques that are more complex. Since our experiments demonstrate that a non-specialist could easily make limestone beads, there is no exclusivity to the craft, and less potential to use it to justify and signal hierarchical positions within a group. Other crafts, such as metalworking, are more plausible candidates, as reinventing the related know-how would be much more costly, and ores are scarce compared to the abundant limestones.

One potential avenue of investigation regarding beadmaking and social stratification is the raw material at some sites. For tubular beads, a marble-like appearance of some beads has been noted (Dieckmann, 1992; Heumüller, 2009). If they

are indeed marble beads, and their distribution in settlements and graves reveals relevant patterns, arguments on social stratification could be made, as the material is less common than limestone. Generally speaking, further work on funerary contexts with beads could reveal patterns of grave good distribution that could indicate social hierarchies, if there are notable contrasting patterns in terms of quality or number of beads in burials. At the current state of research, the absence of representational burials (Gronenborn *et al.*, 2018) and the overall number and preservation of Middle and Late Neolithic graves do not provide a good basis for statements on social structures. Zirngibl (2025) does consider the burials with numerous bead grave goods as status indicators, but interestingly, he points to a broader trend regarding age and suggests that the beads were accumulated over a person's lifetime. Of course, the beads could signify a number of things, including status and skill (see Heumüller, 2009, 2012 for a review). In that context, while still markers of social differentiation, one would have to take a different viewpoint than previous interpretations based in the framework of emerging elites and an economy of surplus production. Certainly, interpretations which consider the beads a form of currency (Maier, 1962; Seidel, 2004) can be rejected when minimal skill is necessary for counterfeiting.

Regarding other economic developments with large-scale implications, our notes on the efficiency of the drills provide support to Heumüller's (2009) criticism of surplus production of limestone beads and provide a basis for controlled experiments on bead output. The surplus argument is based on experiments that suggest one drill yields an average of 4.4 tubular beads, at 2200 rotations per minute, and a 0.86 mm of perforation depth per minute (Seeberger, 1992). Our numbers should not be directly compared to these, as there are too many uncontrolled variables. Highly controlled experiments could provide such comparative data. However, the perceived drop in efficiency of the drills after one or two disc-beads indicates that the estimate of one drill for 4.4 tubular beads is far too high. It should also be noted that the images of the experimental drills published by Seeberger show an intense degree of tip rounding, the likes of which were not observed in the Kohlhau-Abri assemblage. In fact, rounding in our experimental material was more intense than in the archaeological examples as well, suggesting that our use times of 30 min may have already exceeded the actual use times of the archaeological drills.

Researching emerging social inequality and its relationship to economic organization is complex and riddled with pitfalls (see Gronenborn, 2017; Risch, 2016). It is thus important to stick close to the available data, and in its absence, remain agnostic about specific modes of economic organization or the vertical stratification in a certain archaeological context. From a theoretical perspective, Risch (2016) emphasizes that research on social inequality in general must also consider other processes besides economic organization, such as socio-economic distribution strategies, the relationship of violence and warfare to power, as well as different types of oppression of groups of people within a society that need to be considered (Jung & Risch, 2016; Risch, 2016). As such, when it comes to economic processes such as specialization, it is essential to consider the theoretical implications of a particular craft—whether it could have been restricted, and how such a restriction could have been enforced.

Conclusion

Our study provides new insights into Dickenbännli drills on multiple fronts. The traceological evidence of the drills, as well as the chemical characterization of a white residue on a drill tip, shows that the drills at Kohlhau-Abri were used to perforate limestone, likely in the context of beadmaking. From an experimental perspective, we were able to show that simple means, techniques, and skills are sufficient to produce limestone disc-beads. As such, no specialization is required as a consequence of particular fine motor skills or previously gained experience. Furthermore, the controlled experiments provide data on the formation of use traces in drills, serving as potential foundations for more controlled experimental work, in addition to serving as references for Middle and Late Neolithic drills as well as drills in other archaeological contexts. Particularly, the primary criteria, which were present in short as well as longer use times, could help examine the frequently large quantities of Late Neolithic drills. Assessing ridge damage and rounding at the tip could serve as a strong indicator for a drill having been used when detailed analysis of a large number of drills is not feasible due to time constraints. Such "likely used/unused" assessments could aid, for example, in identifying either potential activity zones or caches of unused drills.

Our results underscore the importance of holistic approaches, combining use wear evidence and residue analysis via Raman spectroscopy to strengthen the argument that the drills were used to work limestone, as the polish and calcite identification complement each other well. Furthermore, our design of the experiments in dialogue with the archaeological material and the broader research questions that have been connected to it has allowed us to tackle the "big question" of specialization and challenge the notion that this economic process may have implications for societal structure beyond organization. Not only was it not necessary to specialize to use small lithic drills to make limestone beads. The low level of skill and experience, and the broad availability of the raw materials required in beadmaking indicate that it would have been extremely difficult to restrict the craft other than bysocial means, for which we do not have any direct or indirect evidence. In this regard, characterizing the technology of the drills and the rest of the lithic assemblage, the detailed analysis of which is forthcoming, further supports the findings of our experiments related to skill and sheds light on the manufacturing process of drills and beads as a whole.

Acknowledgements We would like to thank two anonymous reviewers for their comments. We thank Rudi Walter for his valuable help in procuring Jurassic chert raw material and making the drills used in both rounds of the experiment. Thomas Beutelspacher provided helpful insights on fieldwork and find processing. Furthermore, we thank Alfred Pawlik for sharing his at the time unpublished results on the Hornstaad-Hörnle assemblage, which we were able to draw on in the experimental design and interpretation. Regarding the analyses, we are grateful to the Competence Center Archaeometry Baden-Württemberg for access to their Raman spectroscope, and the Material Culture Laboratory for access to the microscopes used in this study. We also thank Nicholas J. Conard for being one of the examiners of the thesis in the context of which a large part of this work was conducted. Natasha Singh kindly proofread for spelling and grammatical errors. Last but not least, a special thanks goes to Robin Andrews, for helping with limestone raw material gathering, and valuable feedback at various points while this study was being conducted.

Author Contributions H.H. : conceptualization, investigation and formal analysis for experiments and all conducted analyses, writing - original draft preparation. C.-J.K. : resources, writing – review & editing. P.S. : investigation and formal analysis regarding Raman spectroscopy, writing - review & editing. Y.T. : conceptualization, resources, writing - review & editing. F.V. : conceptualization, supervision, writing - review & editing.

Funding Open Access funding enabled and organized by Projekt DEAL.

Data Availability All data supporting the findings of this study are available within the paper.

Declarations

Competing Interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Altorfer, K., Leuzinger, U., & Schlichtherle, H. (2016). Doch nicht gleich. Rangabzeichen und Statussymbole. *Begleitband zur Großen Landesausstellung Baden-Württemberg 2016 - 4000 Jahre Pfahlbauten.* Jan Thorbecke Verlag.

Aoyama, K. (1995). Microwear analysis in the Southeast Maya Lowlands: Two case studies at Copan, Honduras. *Latin American Antiquity, 6*(2), 129–144. https://doi.org/10.2307/972148

Bains, R. K. (2012). *The social significance of Neolithic stone bead technologies at Çatalhöyük* UCL (University College London)].

Baysal, E. (2013). Will the real specialist please stand up? Characterising early craft specialisation, a comparative approach for Neolithic Anatolia. *Documenta Praehistorica, 40*, 233–246. https://doi.org/10.4312/dp.40.19

Bonnardin, S. (2004). *La parure funéraire du néolithique ancien en bassins parisien et rhénan: matériaux, techniques, fonctions et usage social* Université Paris 1 Panthéon-Sorbonne].

Bouchard, M., & Smith, D. C. (2003). Catalogue of 45 reference Raman spectra of minerals concerning research in art history or archaeology, especially on corroded metals and coloured glass. *Spectrochimica Acta, 59*, 2247–2266. https://doi.org/10.1016/S1386-1425(03)00069-6

Calley, S., & Grace, R. (1988). Technology and function of micro-borers from Kumartepe (Turkey). *Industries Lithiques: Tracéologie Et Technologie, 1*, 69–81.

Childe, V. G. (1950). The urban revolution. *Town Planning Review, 21*(1), 3–17.

Claud, É., Thiébaut, C., Coudenneau, A., Deschamps, M., Mourre, V., Brenet, M., Chacón-Navarro, M. G., Colonge, D., Lemorini, C., Maury, S., Servelle, C., & Venditti, F. (2019). Stone tool reference collection. *Palethnologie*, *10*.

Coşkunsu, G. (2009). Hole-making tools of Mezraa Teleilat with special attention to micro-borers and cylindrical polished drills and bead production. *NEO-LITHICS, 1*(08), 25.

d'Aujourd'hui, R. (1976). *Bedeutung und Funktion der Dickenbännlispitzen : Mikroskopische Untersuchungen zur Funktionsdeutung von Silexgeräten*. Birkhäuser.

Dieckmann, B. (1992). Die neolithischen Ufersiedlungen von Hornstaad-Hörnle am westlichen Bodensee. In A. Aspes (Ed.), *Symposium Settlement Patterns - Between the Alps and the Black Sea 5th to 2nd millenium B.C.* (Vol. 2). Sezione scienze dell'Uomo.

Dieckmann, B., Hoffstadt, J., Maier, U., & Spatz, H. (1998). Zum Stand der Ausgrabungen auf den "Offwiesen" in Singen, Kreis Konstanz. *Archäologische Ausgrabungen Baden-Württemberg*, 43–47.

Dufresne, W. J. B., Rufledt, C. J., & Marshall, C. P. (2018). Raman spectroscopy of the eight natural carbonate minerals of calcite structure. *Journal of Raman Spectroscopy, 49*(12), 1999–2007. https://doi.org/10.1002/jrs.5481

Engelhardt, B., & Binsteiner, A. (1988). Neolithischer Feuersteinabbau von Arnhofen. *Germania, 66*(1), 1–28.

Ewald, J., & Sedelmaier, J. (1994). Neue Forschungen zum Neolithikum im Kanton Basel-Landschaft. *Jahrbuch der Schweizerischen Gesellschaft für Ur- und Frühgeschichte, 77*, 130–134.

Favre, P., & Jacomet, S. (1998). Branch wood from the lake shore settlements of Horgen Scheller, Switzerland: Evidence for economic specialization in the late neolithic period. *Vegetation History and Archaeobotany, 7*(3), 167–178. https://doi.org/10.1007/BF01374005

Fischer, A., Hansen, P. V., & Rasmussen, P. (1984). Macro and micro wear traces on lithic projectile points. *Journal of Danish Archaeology, 3*(1), 19–46. https://doi.org/10.1080/0108464X.1984.10589910

Garate, D., Rivero, O., Rios-Garaizar, J., Medina-Alcaide, M. Á., Arriolabengoa, M., Intxaurbe, I., Ruiz-López, J. F., Marín-Arroyo, A. B., Rofes, J., García Bustos, P., Torres, A., & Salazar, S. (2023). Unravelling the skills and motivations of Magdalenian artists in the depths of Atxurra Cave (Northern Spain). *Scientific Reports, 13*(1), Article 17340. https://doi.org/10.1038/s41598-023-44520-w

González-Urquijo, J. E., & Ibáñez-Estévez, J. J. (2003). The quantification of use-wear polish using image analysis. First results. *Journal of Archaeological Science, 30*(4), 481–489. https://doi.org/10.1006/jasc.2002.0855

Gronenborn, D. (2017). Some thoughts on political differentiation in Early to Young Neolithic societies in western central Europe. In H. Meller, H.-P. Hahn, R. Jung, & R. Risch (Eds.), *Rich and poor - Competing for resources in prehistoric societies. 8th Archaeological Conference of Central Germany* (Vol. 14/1). Landesamt für Denkmalpflege und Archäologie Sachsen-Anhalt, Landesmuseum für Vorgeschichte.

Gronenborn, D., Strien, H.-C., van Dick, R., & Turchin, P. (2018). Social diversity, social identity, and the emergence of surplus in the Western Central European Neolithic. In H. Meller, D. Gronenborn, & R. Risch (Eds.), *Surplus without the State - Political Forms in Prehistory. 10th Archaeological Conference of Central Germany* (Vol. 18). Landesamt für Denkmalpflege und Archäologie Sachsen-Anhalt, Landesmuseum für Vorgeschichte.

Gurova, M., Bonsall, C., Bradley, B., Anastasova, E., & Cura, P. (2014). An experimental approach to prehistoric drilling and bead manufacturing. *Technology and Experimentation in Archaeology* (pp. 47–56). Archaeopress.

Hansen, S. (2018). Arbeitsteilung, soziale Ungleichheit und Surplus in der Kupferzeit an der unteren Donau 4600–4300 v.Chr. In D. Gronenborn, H.-C. Strien, R. van Dick, & P. Turchin (Eds.), *Surplus without the State - Political Forms in Prehistory. 10th Archaeological Conference of Central Germany* (Vol. 18). Landesamt für Denkmalpflege und Archäologie Sachsen-Anhalt, Landesmuseum für Vorgeschichte.

Heumüller, M. (2009). *Der Schmuck der jungneolithischen Seeufersiedlung Hornstaad-Hörnle IA im Rahmen des mitteleuropäischen Mittel- und Jungneolithikums. Siedlungsarchäologie im Alpenvorland X*. Konrad Theiss Verlag.

Heumüller, M. (2012). Schmuck als Zeichen weit gespannter und lang andauernder Kommunikationsräume im 5. und frühen 4. Jahrtausend. In R. Gleser & V. Becker (Eds.), *Mitteleuropa im 5. Jahrtausend vor Christus. Beiträge zur internationalen Konferenz in Münster 2010. Neolithikum und ältere Metallzeiten 1* (pp. 359–387). LIT Verlag Münster.

Ho Ho Committee. (1979). The Ho Ho classification and nomenclature committee report. In B. Hayden (Ed.), *Lithic use-wear analysis*. Academic Press, Inc.

Hoffstadt, J. (2005). *Siedlungsarchäologie im Alpenvorland VII - Die Untersuchung der Silexartefakte aus der Ufersiedlung Hornstaad-Hörnle IA* (Vol. 90). Konrad Theiss Verlag.

Hoffstadt, J. (2012). Dickenbännlibohrer. In H. Floss (Ed.), *Steinartefakte vom Altpaläolithikum bis in die Neuzeit*. Kerns Verlag.

Jung, R., & Risch, R. (2016). Why are we concerned with social inequality? In H. Meller, H.-P. Hahn, R. Jung, & R. Risch (Eds.), *Rich and poor - Competing for resources in prehistoric societies. 8th Archaeological Conference of Central Germany* (Vol. 14/1). Landesamt für Denkmalpflege und Archäologie Sachsen-Anhalt, Landesmuseum für Vorgeschichte.

Keeley, L. H. (1980). *Experimental Determination of Stone tool Uses: A Microwear Analysis*. The University of Chicago Press.

Kerner, S. (2010). Craft specialisation and its relation with social organisation in the late 6th to early 4th millennium BCE of the Southern Levant. *Paléorient, 36*(1), 179–198.

Kind, C.-J. (1989). *Ulm-Eggingen - Die Ausgrabungen 1982 bis 1985 in der bandkeramischen Siedlung und der mittelalterlichen Wüstung* (Vol. 34). Konrad Theiss Verlag.

Kind, C.-J., & Beutelspacher, T. (2020). Das Kohlhau-Abri: Eine neue Felsdach-Fundstelle in Nachbarschaft zum Lonetal: Ein Vorbericht. *Fundberichte Aus Baden-Württemberg, 40*, 103–124.

Klimscha, F. (2020). New research on the "Kings of Metal": Systems of social distinction in the Copper Age of Southeastern Europe. *Journal of Archaeological Research, 28*(1), 99–138. https://doi.org/10.1007/s10814-019-09132-x

Krauß, R., Bofinger, J., & Weninger, B. (2020). LBK & Vinča – Formation and transformation of Early Neolithic lifestyles in Europe in the second half of the 6th millennium BC. *Quaternary International, 560*, 1–4. https://doi.org/10.1016/j.quaint.2020.05.015

Lafuente, B., Downs, R. T., Yang, H., & Stone, N. (2015). The power of databases: The RRUFF project. In T. Armbruster & R. M. Danisi (Eds.), *Highlights in Mineralogical Crystallography* (pp. 1–30). De Gruyter.

López-Tascón, C., Pedergnana, A., Ollé, A., Rasilla, M., & Mazo, C. (2020). Characterization of the use-wear and residues resulting from limestone working. Experimental approach to the parietal art of La Viña rock shelter (La Manzaneda, Asturias, Spain). *Quaternary International, 569*, 212–227. https://doi.org/10.1016/j.quaint.2020.07.012

Lüning, J. (1988). Frühe Bauern in Mitteleuropa im 6. und 5. Jahrtausend v. Chr. *Jahrbuch Des Römisch-Germanischen Zentralmuseums Mainz, 35*(1), 27–93. https://doi.org/10.11588/jrgzm.1988.1.83144

Lüning, J., & Zürn, H. (1977). *Die Schussenrieder Siedlung im "Schlößlesfeld", Markung Ludwigsburg*. Müller&Gräff.

Maier, R. (1962). Steinröhrenperlen und Kieselanhänger des nordwestalpinen Äneolithikums. *Germania, 40*, 33–43.

Marreiros, J., Calandra, I., Gneisinger, W., Paixão, E., Pedergnana, A., & Schunk, L. (2020). Rethinking use-wear analysis and experimentation as applied to the study of past hominin tool use. *Journal of Paleolithic Archaeology, 3*(3), 475–502. https://doi.org/10.1007/s41982-020-00058-1

Marx, K. (1867). *Das Kapital: Kritik der politischen Oekonomie. Vol. 1: Der Produktionsprozess des Kapitals*. Verlag von Otto Meißner.

Marx, K. (1891). Kritik des Gothaer Programms. *Die Neue Zeit. Revue Des Geistigen Und Öffentlichen Lebens, 18*(1), 561–575.

Müller-Beck, H. (1991). Die Holzartefakte. In H. T. Waterbolk & W. van Zeist (Eds.), *Niederwil, eine Siedlung der Pfyner Kultur*. Verlag Paul Haupt Bern und Stuttgart.

Nadler, M. (1985). Beobachtungen zur Herstellung mittelneolithischer Scheibenperlen. *Natur und Mensch. Jahresmitteilungen der naturhistorischen Gesellschaft Nürnberg e.V.*, 129–133.

Odell, G. H., & Odell-Vereecken, F. (1980). Verifying the reliability of lithic use-wear assessments by 'blind tests': The low-power approach. *Journal of Field Archaeology, 7*(1), 87–120. https://doi.org/10.1179/009346980791505545

Pawlik, A. (2000). Die funktionale Gliederung von Siedlungsarealen im Jungneolithikum: Untersuchungen zum technologischen und soziokulturellen Wandel mit den Methoden der Gebrauchsspurenanalyse. *Zenodo*. https://doi.org/10.5281/zenodo.17541785

Piller, C. K. (2025). Kalksteinperlen im Bayerischen Mittelneolithikum – Zum aktuellen Stand der Forschung. In L. Husty, T. Link, & J. Pechtl (Eds.), *Neue Materialien des Bayerischen Neolithikums 5. Tagung im Kloster Windberg vom 17. bis 19. November 2023* (Vol. 10, pp. 111–138). Würzburg University Press.

Plisson, H. (1985). *Étude fonctionnelle d'outillages lithiques préhistoriques par I'analyse des micro-usures: Recherche methodologique et archéologique* Universitée de Paris 1 Pantheon Sorbonne].

Renfrew, C. (1978). Varna and the social context of early metallurgy. *Antiquity, 52*(206), 199–203. https://doi.org/10.1017/S0003598X00072197

Risch, R. (2016). How did wealth turn into surplus profit? From affluence to scarcity in prehistoric economies. In H. Meller, H.-P. Hahn, R. Jung, & R. Risch (Eds.), *Rich and poor - Competing for resources in prehistoric societies. 8th Archaeological Conference of Central Germany* (Vol. 14/1). Landesamt für Denkmalpflege und Archäologie Sachsen-Anhalt, Landesmuseum für Vorgeschichte.

Roth, G. (2008). *GEBEN UND NEHMEN. Eine wirtschaftshistorische Studie zum neolithischen Hornsteinbergbau von Abensberg-Arnhofen, Kr. Kelheim (Niederbayern) [in IV Bänden]. BAND I: Bergbau* [Dissertation, Universität Köln]. Köln.

Rots, V. (2004). Prehensile wear on flint tools. *Lithic Technology, 29*(1), 7–32. https://doi.org/10.1080/01977261.2004.11721011

Rots, V. (2008). Hafting traces on flint tools. In L. Longo & N. Skakun (Eds.), *"Prehistoric Technology" 40 years later: Functional studies and the Russian Legacy*. Archaeopress

Rousseau, J.-J. (1754). Discours sur l'origine et les fondements de l'inégalité parmi les hommes. In J.-M. Tremblay (Ed.), *Jean-Jacques Rousseau (1754), Discours sur l'origine et les fondements de l'inégalité parmi les hommes. Une édition électronique réalisée à partir du livre de Jean-Jacques Rousseau, Discours sur l'origine et les fondements del'inégalité parmi les hommes.* . Chicoutimi.

Schlichtherle, H. (1990). *Siedlungsarchäologie im Alpenvorland I. Die Sondagen 1973–1978 in den Ufersiedlungen Hornstaad-Hörnle I. Befunde und Funde zum frühen Jungneolithikum am westlichen Bodensee* (Vol. 36). Konrad Theiss Verlag.

Schmidt, P., Bellot-Gurlet, L., Slodczyk, A., & Fröhlich, F. (2012). A hitherto unrecognised band in the Raman spectra of silica rocks: Influence of hydroxylated Si–O bonds (silanole) on the Raman moganite band in chalcedony and flint (SiO2). *Physics and Chemistry of Minerals, 39*(6), 455–464. https://doi.org/10.1007/s00269-012-0499-7

Seeberger, F. (1992). Zur Herstellung der neolithischen Kalkröhrenperlen und Kettenschieber. *Archäologisches Korrespondenzblatt, 22*, 41–45.

Seidel, U. (2004). *Die jungneolithischen Siedlungen von Leonberg-Höfingen, Kr. Böblingen* (Vol. 69). Konrad Theiss.

Semenov, S. A. (1964). *Prehistoric technology : an experimental study of the oldest tools and artefacts from traces of manufacture and wear / S. A. Semenov; transl. and with a pref. by M. W. Thompson* (M. W. Thompson, Trans.). Cory, Adams & Mackay.

Speck, J. (1988). Ein Robbeil aus Aphanit aus der Ufersiedlung Cham ZG, St. Andreas. *Helvetica Archaeologica, 19*, 89–100.

Thomas, J. (2009). Approaching Specialisation: Craft production in Late Neolithic/Copper Age Iberia. *Papers from the Institute of Archaeology, 19*, 67–84. https://doi.org/10.5334/pia.325

Tringham, R., Cooper, G., Odell, G., Voytek, B., & Whitman, A. (1974). Experimentation in the formation of edge damage: A new approach to lithic analysis. *Journal of Field Archaeology, 1*(1–2), 171–196. https://doi.org/10.1179/jfa.1974.1.1-2.171

Van Gijn, A. L. (1990). *The wear and tear of flint: principles of functional analysis applied to Dutch Neolithic assemblages. Analecta Praehistorica Leidensia* Universiteit Leiden]. Leiden.

Vaquero, M., & Romagnoli, F. (2018). Searching for lazy people: The significance of expedient behavior in the interpretation of Paleolithic assemblages. *Journal of Archaeological Method and Theory, 25*(2), 334–367. https://doi.org/10.1007/s10816-017-9339-x

Viel, L. (2021). *Entre richesses et objets ordinaires, les parures de la fin du Néolithique en Provence : approches morphométrique et technique* Aix-Marseille Université].

Viola, S., Bernabo' Brea, M. A., Delcaro, D., Gonzato, F., Longhi, C., Gaj, G., Macellari, R., Salzani, L., Serges, A., Tirabassi, I., & Besse, M. (2017). Types and gesture. The jewellery of the Copper age in the Alps in a techno-typological study. In J. Guilaine & M. Besse (Eds.), *Materials, Productions, Exchange Network and their Impact on the Societies of Neolithic Europe : Proceedings of the XVII UISPP World Congress (1–7 September 2014, Burgos, Spain)*. Archaeopress. https://digital.casalini.it/9781784915254

Weinig, J. (1987). Eine neolithische Schmuckwerkstätte aus Gaimersheim. In *Das archäologische Jahr in Bayern*. Konrad Theiss Verlag GmbH & Co.

Willms, C. (1982). *Zwei Fundplätze der Michelsberger Kultur aus dem westlichen Münsterland, gleichzeitig ein Beitrag zum neolithischen Silexhandel in Mitteleuropa*. Verlag August Lax.

Wright, K. I. (2014). Domestication and inequality? Households, corporate groups and food processing tools at Neolithic Çatalhöyük. *Journal of Anthropological Archaeology, 33*, 1–33. https://doi.org/10.1016/j.jaa.2013.09.007

Wright, K. I., Critchley, P., Garrard, A., Baird, D., Bains, R., & Groom, S. (2008). Stone Bead Technologies and Early Craft Specialization: Insights from Two Neolithic Sites in Eastern Jordan. *Levant, 40*(2), 131–165. https://doi.org/10.1179/175638008X348016

Yamada, S., & Sawada, A. (1993). The method of description for polished surfaces. In P. C. Anderson, S. Beyries, M. Otte, & H. Plisson (Eds.), *Traces et fonction: Les gestes retrouvés* (Vol. 2, pp. 447–458). Centre de Recherches Archéologiques du CNRS.

Zirngibl, B. (2025). Naturwissenschaftliche und sozialarchäologische Aspekte zu einer mittelneolithischen Frauenbestattung mit zahlreichen Kalksteinperlen vom Altdorfer „Hochrückacker", Lkr. Landshut sowie Einschätzungen zur Trageweise zeitgleichen Perlenschmucks. In L. Husty, T. Link, & J. Pechtl

(Eds.), *Neue Materialien des Bayerischen Neolithikums 5. Tagung im Kloster Windberg vom 17. bis 19. November 2023* (Vol. 10, pp. 139–159). Würzburg University Press.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Hannah Huber[1,2] · Claus-Joachim Kind[4] · Patrick Schmidt[1,3] · Yvonne Tafelmaier[4] · Flavia Venditti[1]

✉ Hannah Huber
hannah.huber@uni-tuebingen.de

✉ Flavia Venditti
flavia.venditti@uni-tuebingen.de

1 Working Group Early Prehistory and Quaternary Ecology, Department of Geosciences, University of Tübingen, Tübingen, Germany

2 Department of Pre- and Protohistory, University of Tübingen, Tübingen, Germany

3 Petrology and Mineral Resources, Department of Geosciences, University of Tübingen, Tübingen, Germany

4 State Office for Cultural Heritage Baden-Württemberg, Tübingen, Germany